

MXN442-N11423579-Siu-Wing-Hei

2024-10-20

```
train <- read.csv("C:/Users/siusa/Downloads/MXN442-N11423579-Siu-Wing-Hei/train.csv")
```

```
test <- read.csv("C:/Users/siusa/Downloads/MXN442-N11423579-Siu-Wing-Hei/test.csv")
```

```
test$Name[415] <- "Oliva y Ocana, Miss. Fermina" # becuase Oliva y Ocana, Dona. Fermina do not have Nick
```

```
gender_submission <- read.csv("C:/Users/siusa/Downloads/MXN442-N11423579-Siu-Wing-Hei/gender_submission.csv")
```

Data processing

```
colnames(train)
```

```
## [1] "PassengerId" "Survived"      "Pclass"        "Name"          "Sex"
## [6] "Age"          "SibSp"         "Parch"         "Ticket"        "Fare"
## [11] "Cabin"        "Embarked"
```

```
str(train)
```

```
## 'data.frame':      891 obs. of  12 variables:
## $ PassengerId: int   1  2  3  4  5  6  7  8  9 10 ...
## $ Survived   : int   0  1  1  1  0  0  0  0  1  1 ...
## $ Pclass     : int   3  1  3  1  3  3  1  3  3  2 ...
## $ Name       : chr   "Braund, Mr. Owen Harris" "Cumings, Mrs. John Bradley (Florence Briggs Thayer)"
## $ Sex        : chr   "male" "female" "female" "female" ...
## $ Age        : num   22  38  26  35  35 NA  54  2  27  14 ...
## $ SibSp      : int   1  1  0  1  0  0  0  3  0  1 ...
## $ Parch      : int   0  0  0  0  0  0  0  1  2  0 ...
## $ Ticket     : chr   "A/5 21171" "PC 17599" "STON/O2. 3101282" "113803" ...
## $ Fare       : num   7.25 71.28 7.92 53.1 8.05 ...
## $ Cabin      : chr   "" "C85" "" "C123" ...
## $ Embarked   : chr   "S" "C" "S" "S" ...
```

```
str(test)
```

```
## 'data.frame':      418 obs. of  11 variables:
## $ PassengerId: int  892 893 894 895 896 897 898 899 900 901 ...
## $ Pclass     : int   3  3  2  3  3  3  2  3  3 ...
## $ Name       : chr   "Kelly, Mr. James" "Wilkes, Mrs. James (Ellen Needs)" "Myles, Mr. Thomas Francis"
```

```
## $ Sex      : chr  "male" "female" "male" "male" ...
## $ Age      : num  34.5 47 62 27 22 14 30 26 18 21 ...
## $ SibSp    : int   0 1 0 0 1 0 0 1 0 2 ...
## $ Parch    : int   0 0 0 0 1 0 0 1 0 0 ...
## $ Ticket   : chr   "330911" "363272" "240276" "315154" ...
## $ Fare     : num   7.83 7 9.69 8.66 12.29 ...
## $ Cabin    : chr    "" "" "" "" ...
## $ Embarked : chr   "Q" "S" "Q" "S" ...
```

```
str(gender_submission)
```

```
## 'data.frame': 418 obs. of 2 variables:
## $ PassengerId: int 892 893 894 895 896 897 898 899 900 901 ...
## $ Survived : int 0 1 0 0 1 0 1 0 1 0 ...
```

```
changedtpye_train<-data.frame(PassengerId = train$PassengerId,
                              Survived = as.factor(train$Survived),
                              Pclass = as.factor(train$Pclass),
                              Name = train$Name,
                              Sex = as.factor(train$Sex),
                              Age = train$Age,
                              SibSp = train$SibSp,
                              Parch = train$Parch,
                              Ticket = train$Ticket,
                              Fare = train$Fare,
                              Cabin = train$Cabin,
                              Embarked = train$Embarked
                              )
```

```
str(changedtpye_train)
```

```
## 'data.frame': 891 obs. of 12 variables:
## $ PassengerId: int 1 2 3 4 5 6 7 8 9 10 ...
## $ Survived : Factor w/ 2 levels "0","1": 1 2 2 2 1 1 1 1 2 2 ...
## $ Pclass : Factor w/ 3 levels "1","2","3": 3 1 3 1 3 3 1 3 3 2 ...
## $ Name : chr "Braund, Mr. Owen Harris" "Cumings, Mrs. John Bradley (Florence Briggs Thayer)"
## $ Sex : Factor w/ 2 levels "female","male": 2 1 1 1 2 2 2 2 1 1 ...
## $ Age : num 22 38 26 35 35 NA 54 2 27 14 ...
## $ SibSp : int 1 1 0 1 0 0 0 3 0 1 ...
## $ Parch : int 0 0 0 0 0 0 0 1 2 0 ...
## $ Ticket : chr "A/5 21171" "PC 17599" "STON/O2. 3101282" "113803" ...
## $ Fare : num 7.25 71.28 7.92 53.1 8.05 ...
## $ Cabin : chr "" "C85" "" "C123" ...
## $ Embarked : chr "S" "C" "S" "S" ...
```

```
changedtpye_test<-data.frame(PassengerId = test$PassengerId,
                              Pclass = as.factor(test$Pclass),
                              Name = test$Name,
                              Sex = as.factor(test$Sex),
                              Age = test$Age,
                              SibSp = test$SibSp,
                              Parch = test$Parch,
```

```

        Ticket = test$Ticket,
        Fare = test$Fare,
        Cabin = test$Cabin,
        Embarked = test$Embarked
    )

str(changedtpye_test)

## 'data.frame':  418 obs. of  11 variables:
## $ PassengerId: int  892 893 894 895 896 897 898 899 900 901 ...
## $ Pclass     : Factor w/ 3 levels "1","2","3": 3 3 2 3 3 3 3 2 3 3 ...
## $ Name       : chr  "Kelly, Mr. James" "Wilkes, Mrs. James (Ellen Needs)" "Myles, Mr. Thomas Francis" ...
## $ Sex        : Factor w/ 2 levels "female","male": 2 1 2 2 1 2 1 2 1 2 ...
## $ Age        : num  34.5 47 62 27 22 14 30 26 18 21 ...
## $ SibSp      : int  0 1 0 0 1 0 0 1 0 2 ...
## $ Parch      : int  0 0 0 0 1 0 0 1 0 0 ...
## $ Ticket     : chr  "330911" "363272" "240276" "315154" ...
## $ Fare       : num  7.83 7 9.69 8.66 12.29 ...
## $ Cabin      : chr  "" "" "" "" ...
## $ Embarked   : chr  "Q" "S" "Q" "S" ...

changedtpye_gender_submission<-data.frame(PassengerId = gender_submission$PassengerId,
                                           Survived = as.factor(gender_submission$Survived)
                                           )

str(changedtpye_gender_submission)

## 'data.frame':  418 obs. of  2 variables:
## $ PassengerId: int  892 893 894 895 896 897 898 899 900 901 ...
## $ Survived   : Factor w/ 2 levels "0","1": 1 2 1 1 2 1 2 1 2 1 ...

process_train <- subset(changedtpye_train, select = -PassengerId)

# change age to mean
train_meanofAge <- mean(process_train$Age, na.rm = TRUE)
process_train$Age[is.na(process_train$Age)] <- train_meanofAge

process_train <- subset(process_train, select = -Ticket)

process_train <- subset(process_train, select = -Fare)

process_train <- subset(process_train, select = -Cabin)

process_train <- subset(process_train, select = -Embarked)

process_train$Nickname <- sub(".*,\\s*(\\w+)\\.\\s*", "\\1", process_train$Name)
process_train$Nickname <- as.factor(process_train$Nickname)

process_train <- subset(process_train, select = -Name)#must do this after already change Nickname

head(process_train)

```

```
##   Survived Pclass   Sex      Age SibSp Parch Nickname
## 1         0      3  male 22.00000    1     0      Mr
## 2         1      1 female 38.00000    1     0     Mrs
## 3         1      3 female 26.00000    0     0    Miss
## 4         1      1 female 35.00000    1     0     Mrs
## 5         0      3  male 35.00000    0     0      Mr
## 6         0      3  male 29.69912    0     0      Mr
```

```
process_test <- subset(changedtpye_test, select = -PassengerId)

# change age to mean
test_meanofAge <- mean(process_test$Age, na.rm = TRUE)
process_test$Age[is.na(process_test$Age)] <- test_meanofAge

process_test <- subset(process_test, select = -Ticket)

process_test <- subset(process_test, select = -Fare)

process_test <- subset(process_test, select = -Cabin)

process_test <- subset(process_test, select = -Embarked)

process_test$Nickname <- sub(".*,\\s*(\\w+)\\.\\s*", "\\1", process_test$Name)
process_test$Nickname <- as.factor(process_test$Nickname)
#process_test$Nickname[415] <- "Miss" # becuae Oliva y Ocana, Dona. Fermina do not have Nickname

process_test <- subset(process_test, select = -Name)#must do this after already change Nickname

head(process_test)
```

```
##   Pclass   Sex  Age SibSp Parch Nickname
## 1      3  male 34.5     0     0      Mr
## 2      3 female 47.0     1     0     Mrs
## 3      2  male 62.0     0     0      Mr
## 4      3  male 27.0     0     0      Mr
## 5      3 female 22.0     1     1     Mrs
## 6      3  male 14.0     0     0      Mr
```

```
library(caret)
```

```
## Loading required package: ggplot2
```

```
## Loading required package: lattice
```

```
library(e1071)
library(randomForest)
```

```
## randomForest 4.7-1.1
```

```
## Type rfNews() to see new features/changes/bug fixes.
```

```
##
## Attaching package: 'randomForest'

## The following object is masked from 'package:ggplot2':
##
##     margin
```

```
library(class)
```

Article Model

My Confusion Matrix Structural

```
##(TP,TN,FP,FN Location is different to the Leture, Week 5: Classification p .26)
```

```
cat("My Confusion Matrix Structural(different to the Leture, Week 5: Classification p .26)")
```

```
## My Confusion Matrix Structural(different to the Leture, Week 5: Classification p .26)
```

```
cat("\n")
```

```
cat("\n")
```

```
cat("Row 1 (0) = pred_value == 0 = Predicted Negative ")
```

```
## Row 1 (0) = pred_value == 0 = Predicted Negative
```

```
cat("\n")
```

```
cat("Row 2 (1) = pred_value == 1 = Predicted Positive ")
```

```
## Row 2 (1) = pred_value == 1 = Predicted Positive
```

```
cat("\n")
```

```
cat("Column (0) = changedtpye_gender_submission$Survived == 0 = Actual Negative")
```

```
## Column (0) = changedtpye_gender_submission$Survived == 0 = Actual Negative
```

```
cat("\n")
```

```
cat("Column (1) = changedtpye_gender_submission$Survived == 1 = Actual Positive")
```

```
## Column (1) = changedtpye_gender_submission$Survived == 1 = Actual Positive
```

```
cat("\n")
```

```
cat("\n")
```

```
cat("\n")
```

```
cat("\n")
```

```
cat("      changedtpye_gender_submission$Survived == 0      changedtpye_gender_submission$Survived == 1")
```

```
##      changedtpye_gender_submission$Survived == 0      changedtpye_gender_submission$Survived == 1
```

```
cat("      Actual Negative      Actual Positive")
```

```
##      Actual Negative      Actual Positive
```

```
cat("\n")
```

```
cat("\n")
```

```
cat("Confusion Matrix      0      1")
```

```
## Confusion Matrix      0      1
```

```
cat("\n")
```

```
cat("\n")
```

```
cat("pred_value == 0,      0      TN      FN")
```

```
## pred_value == 0,      0      TN      FN
```

```
cat("\n")
```

```
cat("Predicted Negative")
```

```
## Predicted Negative
```

```
cat("\n")
```

```
cat("\n")
```

```
cat("pred_value == 1,      1      FP      TP")
```

```
## pred_value == 1,      1      FP      TP
```

```
cat("\n")
```

```
cat("Predicted Positive ")
```

```
## Predicted Positive
```

Logistic

Logistic with out cross-validation

```
process_test$Sex <- factor(process_test$Sex, levels = levels(process_train$Sex))
process_test$Nickname <- factor(process_test$Nickname, levels = levels(process_train$Nickname))
set.seed(1)
logisticwithoutcross <- glm(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,
                             data = process_train,
                             family = binomial)

pred_logisticwithoutcross <- predict(logisticwithoutcross, newdata = process_test)

pred_value_logisticwithoutcross <- ifelse(pred_logisticwithoutcross > 0.5, 1, 0)

conmatrix_logisticwithoutcross <- table(pred_value_logisticwithoutcross, changedtpye_gender_submission$Survived)
print(conmatrix_logisticwithoutcross)

##
## pred_value_logisticwithoutcross    0    1
##                                0 253  18
##                                1  13 134

TP_logisticwithoutcross <- sum(pred_value_logisticwithoutcross == 1 & changedtpye_gender_submission$Survived == 1)
TN_logisticwithoutcross <- sum(pred_value_logisticwithoutcross == 0 & changedtpye_gender_submission$Survived == 0)
FP_logisticwithoutcross <- sum(pred_value_logisticwithoutcross == 1 & changedtpye_gender_submission$Survived == 0)
FN_logisticwithoutcross <- sum(pred_value_logisticwithoutcross == 0 & changedtpye_gender_submission$Survived == 1)

cat("True Positives in Titanic Logistic (TP):", TP_logisticwithoutcross, "\n")

## True Positives in Titanic Logistic (TP): 134

cat("True Negatives in Titanic Logistic (TN):", TN_logisticwithoutcross, "\n")
```

```
## True Negatives in Titanic Logistic (TN): 253
```

```
cat("False Positives in Titanic Logistic (FP):", FP_logisticwithoutcross, "\n")
```

```
## False Positives in Titanic Logistic (FP): 13
```

```
cat("False Negatives in Titanic Logistic (FN):", FN_logisticwithoutcross, "\n")
```

```
## False Negatives in Titanic Logistic (FN): 18
```

```
acc_logisticwithoutcross <- (TP_logisticwithoutcross + TN_logisticwithoutcross) / (TP_logisticwithoutcross + FN_logisticwithoutcross + FP_logisticwithoutcross)
```

```
precision_logisticwithoutcross <- (TP_logisticwithoutcross) / (TP_logisticwithoutcross + FP_logisticwithoutcross)
```

```
reca_logisticwithoutcross <- (TP_logisticwithoutcross) / (TP_logisticwithoutcross + FN_logisticwithoutcross)
```

```
f1_logisticwithoutcross <- 2 * ((precision_logisticwithoutcross * reca_logisticwithoutcross) / (precision_logisticwithoutcross + reca_logisticwithoutcross))
```

```
cat("Accuracy of Titanic Logistic:", acc_logisticwithoutcross, "\n")
```

```
## Accuracy of Titanic Logistic: 0.9258373
```

```
cat("Precision of Titanic Logistic:", precision_logisticwithoutcross, "\n")
```

```
## Precision of Titanic Logistic: 0.9115646
```

```
cat("Recall of Titanic Logistic:", reca_logisticwithoutcross, "\n")
```

```
## Recall of Titanic Logistic: 0.8815789
```

```
cat("F1-Score of Titanic Logistic:", f1_logisticwithoutcross, "\n")
```

```
## F1-Score of Titanic Logistic: 0.8963211
```

Logistic with k-fold cross-validation

Logistic with k-fold cross-validation (k=10)

```
process_test$Sex <- factor(process_test$Sex, levels = levels(process_train$Sex))
process_test$Nickname <- factor(process_test$Nickname, levels = levels(process_train$Nickname))

set.seed(1)
k10 <- trainControl(method = "cv", number = 10)

logistic10 <- train(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,
                    data = process_train,
                    method = "glm",
                    family = binomial,
                    trControl = k10)
```

```

## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

## Warning in predict.lm(object, newdata, se.fit, scale = 1, type = if (type == :
## prediction from rank-deficient fit; attr(*, "non-estim") has doubtful cases
## Warning in predict.lm(object, newdata, se.fit, scale = 1, type = if (type == :
## prediction from rank-deficient fit; attr(*, "non-estim") has doubtful cases
## Warning in predict.lm(object, newdata, se.fit, scale = 1, type = if (type == :
## prediction from rank-deficient fit; attr(*, "non-estim") has doubtful cases

## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

## Warning in predict.lm(object, newdata, se.fit, scale = 1, type = if (type == :
## prediction from rank-deficient fit; attr(*, "non-estim") has doubtful cases
## Warning in predict.lm(object, newdata, se.fit, scale = 1, type = if (type == :
## prediction from rank-deficient fit; attr(*, "non-estim") has doubtful cases
## Warning in predict.lm(object, newdata, se.fit, scale = 1, type = if (type == :
## prediction from rank-deficient fit; attr(*, "non-estim") has doubtful cases

pred_logistick10 <- predict(logistick10, newdata = process_test)

pred_value_logistick10 <- pred_logistick10

conmatrix_logistick10 <- table(pred_value_logistick10, changedtpye_gender_submission$Survived)

print(conmatrix_logistick10)

##
## pred_value_logistick10    0    1
##                0 247    6
##                1  19 146

TP_logistick10 <- sum(pred_value_logistick10 == 1 & changedtpye_gender_submission$Survived == 1)
TN_logistick10 <- sum(pred_value_logistick10 == 0 & changedtpye_gender_submission$Survived == 0)
FP_logistick10 <- sum(pred_value_logistick10 == 1 & changedtpye_gender_submission$Survived == 0)
FN_logistick10 <- sum(pred_value_logistick10 == 0 & changedtpye_gender_submission$Survived == 1)

cat("True Positives in Titanic Logistic (TP):", TP_logistick10, "\n")

## True Positives in Titanic Logistic (TP): 146

```

```

cat("True Negatives in Titanic Logistic (TN):", TN_logistick10, "\n")

## True Negatives in Titanic Logistic (TN): 247

cat("False Positives in Titanic Logistic (FP):", FP_logistick10, "\n")

## False Positives in Titanic Logistic (FP): 19

cat("False Negatives in Titanic Logistic (FN):", FN_logistick10, "\n")

## False Negatives in Titanic Logistic (FN): 6

acc_logistick10 <- (TP_logistick10 + TN_logistick10) / (TP_logistick10 + TN_logistick10 + FP_logistick10)
precision_logistick10 <- (TP_logistick10) / (TP_logistick10 + FP_logistick10)
reca_logistick10 <- (TP_logistick10) / (TP_logistick10 + FN_logistick10)
f1_logistick10 <- 2 * ((precision_logistick10 * reca_logistick10) / (precision_logistick10 + reca_logistick10))

cat("Accuracy of Titanic Logistic:", acc_logistick10, "\n")

## Accuracy of Titanic Logistic: 0.9401914

cat("Precision of Titanic Logistic:", precision_logistick10, "\n")

## Precision of Titanic Logistic: 0.8848485

cat("Recall of Titanic Logistic:", reca_logistick10, "\n")

## Recall of Titanic Logistic: 0.9605263

cat("F1-Score of Titanic Logistic:", f1_logistick10, "\n")

## F1-Score of Titanic Logistic: 0.9211356

```

Logistic with k-fold cross-validation (k=5)

```

process_test$Sex <- factor(process_test$Sex, levels = levels(process_train$Sex))
process_test$Nickname <- factor(process_test$Nickname, levels = levels(process_train$Nickname))

set.seed(1)
k5 <- trainControl(method = "cv", number = 5)

logistick5 <- train(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,
                    data = process_train,
                    method = "glm",
                    family = binomial,
                    trControl = k5)

```

```
## Warning: glm.fit: algorithm did not converge
```

```
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

```
## Warning in predict.lm(object, newdata, se.fit, scale = 1, type = if (type == :  
## prediction from rank-deficient fit; attr(*, "non-estim") has doubtful cases  
## Warning in predict.lm(object, newdata, se.fit, scale = 1, type = if (type == :  
## prediction from rank-deficient fit; attr(*, "non-estim") has doubtful cases  
## Warning in predict.lm(object, newdata, se.fit, scale = 1, type = if (type == :  
## prediction from rank-deficient fit; attr(*, "non-estim") has doubtful cases  
## Warning in predict.lm(object, newdata, se.fit, scale = 1, type = if (type == :  
## prediction from rank-deficient fit; attr(*, "non-estim") has doubtful cases
```

```
pred_logistick5 <- predict(logistick5, newdata = process_test)
```

```
pred_value_logistick5 <- pred_logistick5
```

```
conmatrix_logistick5 <- table(pred_value_logistick5, changedtpye_gender_submission$Survived)  
print(conmatrix_logistick5)
```

```
##  
## pred_value_logistick5    0    1  
##                0 247    6  
##                1  19 146
```

```
TP_logistick5 <- sum(pred_value_logistick5 == 1 & changedtpye_gender_submission$Survived == 1)  
TN_logistick5 <- sum(pred_value_logistick5 == 0 & changedtpye_gender_submission$Survived == 0)  
FP_logistick5 <- sum(pred_value_logistick5 == 1 & changedtpye_gender_submission$Survived == 0)  
FN_logistick5 <- sum(pred_value_logistick5 == 0 & changedtpye_gender_submission$Survived == 1)  
cat("True Positives in Titanic Logistic (TP):", TP_logistick5, "\n")
```

```
## True Positives in Titanic Logistic (TP): 146
```

```
cat("True Negatives in Titanic Logistic (TN):", TN_logistick5, "\n")
```

```
## True Negatives in Titanic Logistic (TN): 247
```

```
cat("False Positives in Titanic Logistic (FP):", FP_logistick5, "\n")
```

```
## False Positives in Titanic Logistic (FP): 19
```

```
cat("False Negatives in Titanic Logistic (FN):", FN_logistick5, "\n")
```

```
## False Negatives in Titanic Logistic (FN): 6
```

```
acc_logistick5 <- (TP_logistick5 + TN_logistick5) / (TP_logistick5 + TN_logistick5 + FP_logistick5 + FN_logistick5)
```

```
precision_logistick5 <- (TP_logistick5) / (TP_logistick5 + FP_logistick5)
```

```
reca_logistick5 <- (TP_logistick5) / (TP_logistick5 + FN_logistick5)
```

```
f1_logistick5 <- 2 * ((precision_logistick5 * reca_logistick5) / (precision_logistick5 + reca_logistick5))
```

```
cat("Accuracy of Titanic Logistic:", acc_logistick5, "\n")
```

```
## Accuracy of Titanic Logistic: 0.9401914
```

```
cat("Precision of Titanic Logistic:", precision_logistick5, "\n")
```

```
## Precision of Titanic Logistic: 0.8848485
```

```
cat("Recall of Titanic Logistic:", reca_logistick5, "\n")
```

```
## Recall of Titanic Logistic: 0.9605263
```

```
cat("F1-Score of Titanic Logistic:", f1_logistick5, "\n")
```

```
## F1-Score of Titanic Logistic: 0.9211356
```

Logistic with k-fold cross-validation (k=4)

```
process_test$Sex <- factor(process_test$Sex, levels = levels(process_train$Sex))
process_test$Nickname <- factor(process_test$Nickname, levels = levels(process_train$Nickname))
```

```
set.seed(1)
```

```
k4 <- trainControl(method = "cv", number = 4)
```

```
logistick4 <- train(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,
                    data = process_train,
                    method = "glm",
                    family = binomial,
                    trControl = k4)
```

```
## Warning in predict.lm(object, newdata, se.fit, scale = 1, type = if (type == :
## prediction from rank-deficient fit; attr(*, "non-estim") has doubtful cases
```

```
## Warning: glm.fit: algorithm did not converge
```

```
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

```
## Warning in predict.lm(object, newdata, se.fit, scale = 1, type = if (type == :  
## prediction from rank-deficient fit; attr(*, "non-estim") has doubtful cases
```

```
## Warning: glm.fit: algorithm did not converge
```

```
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

```
## Warning in predict.lm(object, newdata, se.fit, scale = 1, type = if (type == :  
## prediction from rank-deficient fit; attr(*, "non-estim") has doubtful cases
```

```
pred_logistick4 <- predict(logistick4, newdata = process_test)
```

```
pred_value_logistick4 <- pred_logistick4
```

```
conmatrix_logistick4 <- table(pred_value_logistick4, changedtpye_gender_submission$Survived)  
print(conmatrix_logistick4)
```

```
##  
## pred_value_logistick4    0    1  
##                0 247    6  
##                1  19 146
```

```
TP_logistick4 <- sum(pred_value_logistick4 == 1 & changedtpye_gender_submission$Survived == 1)
```

```
TN_logistick4 <- sum(pred_value_logistick4 == 0 & changedtpye_gender_submission$Survived == 0)
```

```
FP_logistick4 <- sum(pred_value_logistick4 == 1 & changedtpye_gender_submission$Survived == 0)
```

```
FN_logistick4 <- sum(pred_value_logistick4 == 0 & changedtpye_gender_submission$Survived == 1)
```

```
cat("True Positives in Titanic Logistic (TP):", TP_logistick4, "\n")
```

```
## True Positives in Titanic Logistic (TP): 146
```

```
cat("True Negatives in Titanic Logistic (TN):", TN_logistick4, "\n")
```

```
## True Negatives in Titanic Logistic (TN): 247
```

```
cat("False Positives in Titanic Logistic (FP):", FP_logistick4, "\n")
```

```
## False Positives in Titanic Logistic (FP): 19
```

```
cat("False Negatives in Titanic Logistic (FN):", FN_logistick4, "\n")
```

```
## False Negatives in Titanic Logistic (FN): 6
```

```

acc_logistick4 <- (TP_logistick4 + TN_logistick4) / (TP_logistick4 + TN_logistick4 + FP_logistick4 + FN_logistick4)
precision_logistick4 <- (TP_logistick4) / (TP_logistick4 + FP_logistick4)
reca_logistick4 <- (TP_logistick4) / (TP_logistick4 + FN_logistick4)
f1_logistick4 <- 2 * ((precision_logistick4 * reca_logistick4) / (precision_logistick4 + reca_logistick4))
cat("Accuracy of Titanic Logistic:", acc_logistick4, "\n")

```

```
## Accuracy of Titanic Logistic: 0.9401914
```

```
cat("Precision of Titanic Logistic:", precision_logistick4, "\n")
```

```
## Precision of Titanic Logistic: 0.8848485
```

```
cat("Recall of Titanic Logistic:", reca_logistick4, "\n")
```

```
## Recall of Titanic Logistic: 0.9605263
```

```
cat("F1-Score of Titanic Logistic:", f1_logistick4, "\n")
```

```
## F1-Score of Titanic Logistic: 0.9211356
```

K nearest neighbors(KNN)

KNN with out cross-validation

```

library(class)
library(caret)
library(kknn)

```

```

##
## Attaching package: 'kknn'

```

```

## The following object is masked from 'package:caret':
##
##      contr.dummy

```

```
library(kknn)
```

```

process_test$Sex <- factor(process_test$Sex, levels = levels(process_train$Sex))
process_test$Nickname <- factor(process_test$Nickname, levels = levels(process_train$Nickname))

```

```

set.seed(1)
KNNwithoutcross <- knn(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname, train = process_train

pre_KNNwithoutcross <- predict(KNNwithoutcross, newdata = process_test)

conmatrix_KNNwithoutcross <- table(pre_KNNwithoutcross , changedtpye_gender_submission$Survived)

print(conmatrix_KNNwithoutcross)

##
## pre_KNNwithoutcross    0    1
##           0 218   41
##           1  48  111

TP_KNNwithoutcross <- sum(pre_KNNwithoutcross == 1 & changedtpye_gender_submission$Survived == 1)
TN_KNNwithoutcross <- sum(pre_KNNwithoutcross == 0 & changedtpye_gender_submission$Survived == 0)
FP_KNNwithoutcross <- sum(pre_KNNwithoutcross == 1 & changedtpye_gender_submission$Survived == 0)
FN_KNNwithoutcross <- sum(pre_KNNwithoutcross == 0 & changedtpye_gender_submission$Survived == 1)

cat("True Positives in Titanic KNN:", TP_KNNwithoutcross, "\n")

## True Positives in Titanic KNN: 111

cat("True Negatives in Titanic KNN:", TN_KNNwithoutcross, "\n")

## True Negatives in Titanic KNN: 218

cat("False Positives in Titanic KNN:", FP_KNNwithoutcross, "\n")

## False Positives in Titanic KNN: 48

cat("False Negatives in Titanic KNN:", FN_KNNwithoutcross, "\n")

## False Negatives in Titanic KNN: 41

acc_KNNwithoutcross <- (TP_KNNwithoutcross + TN_KNNwithoutcross) / (TP_KNNwithoutcross + TN_KNNwithoutcross)
precision_KNNwithoutcross <- (TP_KNNwithoutcross) / (TP_KNNwithoutcross + FP_KNNwithoutcross)
reca_KNNwithoutcross <- (TP_KNNwithoutcross) / (TP_KNNwithoutcross + FN_KNNwithoutcross)
f1_KNNwithoutcross <- 2 * ((precision_KNNwithoutcross * reca_KNNwithoutcross) / (precision_KNNwithoutcross + reca_KNNwithoutcross))

cat("Accuracy of Titanic KNN:", acc_KNNwithoutcross, "\n")

```

```
## Accuracy of Titanic KNN: 0.7870813
```

```
cat("Precision of Titanic KNN:", precision_KNNwithoutcross, "\n")
```

```
## Precision of Titanic KNN: 0.6981132
```

```
cat("Recall of Titanic KNN:", reca_KNNwithoutcross, "\n")
```

```
## Recall of Titanic KNN: 0.7302632
```

```
cat("F1-Score of Titanic KNN:", f1_KNNwithoutcross, "\n")
```

```
## F1-Score of Titanic KNN: 0.7138264
```

KNN with k-fold cross-validation (k=10)

```
library(kknn)
```

```
process_test$Sex <- factor(process_test$Sex, levels = levels(process_train$Sex))  
process_test$Nickname <- factor(process_test$Nickname, levels = levels(process_train$Nickname))
```

```
set.seed(1)
```

```
k10 <- trainControl(method = "cv", number = 10)
```

```
KNNk10 <- train(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,  
               data = process_train, trControl = k10, method = "kknn",  
               tuneGrid = expand.grid(kmax = 5, distance = 2, kernel = "optimal"))
```

```
pre_KNNk10 <- predict(KNNk10, newdata = process_test)
```

```
conmatrix_KNNk10 <- table(pre_KNNk10, changedtpye_gender_submission$Survived)
```

```
print(conmatrix_KNNk10)
```

```
##
```

```
## pre_KNNk10    0    1
```

```
##           0 223  42
```

```
##           1  43 110
```

```
TP_KNNk10 <- sum(pre_KNNk10 == 1 & changedtpye_gender_submission$Survived == 1)
```

```
TN_KNNk10 <- sum(pre_KNNk10 == 0 & changedtpye_gender_submission$Survived == 0)
```

```
FP_KNNk10 <- sum(pre_KNNk10 == 1 & changedtpye_gender_submission$Survived == 0)
```

```
FN_KNNk10 <- sum(pre_KNNk10 == 0 & changedtpye_gender_submission$Survived == 1)
```

```
cat("True Positives in Titanic KNN:", TP_KNNk10, "\n")
```

```
## True Positives in Titanic KNN: 110
```

```
cat("True Negatives in Titanic KNN:", TN_KNNk10, "\n")
```

```
## True Negatives in Titanic KNN: 223
```

```
cat("False Positives in Titanic KNN:", FP_KNNk10, "\n")
```

```
## False Positives in Titanic KNN: 43
```

```
cat("False Negatives in Titanic KNN:", FN_KNNk10, "\n")
```

```
## False Negatives in Titanic KNN: 42
```

```
acc_KNNk10 <- (TP_KNNk10 + TN_KNNk10) / (TP_KNNk10 + TN_KNNk10 + FP_KNNk10 + FN_KNNk10)
```

```
precision_KNNk10 <- (TP_KNNk10) / (TP_KNNk10 + FP_KNNk10)
```

```
reca_KNNk10 <- (TP_KNNk10) / (TP_KNNk10 + FN_KNNk10)
```

```
f1_KNNk10 <- 2 * ((precision_KNNk10 * reca_KNNk10) / (precision_KNNk10 + reca_KNNk10))
```

```
cat("Accuracy of Titanic KNN:", acc_KNNk10, "\n")
```

```
## Accuracy of Titanic KNN: 0.7966507
```

```
cat("Precision of Titanic KNN:", precision_KNNk10, "\n")
```

```
## Precision of Titanic KNN: 0.7189542
```

```
cat("Recall of Titanic KNN:", reca_KNNk10, "\n")
```

```
## Recall of Titanic KNN: 0.7236842
```

```
cat("F1-Score of Titanic KNN:", f1_KNNk10, "\n")
```

```
## F1-Score of Titanic KNN: 0.7213115
```

KNN with k-fold cross-validation (k=5)

```
library(kknn)
```

```
process_test$Sex <- factor(process_test$Sex, levels = levels(process_train$Sex))
```

```
process_test$Nickname <- factor(process_test$Nickname, levels = levels(process_train$Nickname))
```

```
set.seed(1)
```

```

k5 <- trainControl(method = "cv", number = 5)

KNNk5 <- train(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,
               data = process_train, trControl = k5, method = "knn",
               tuneGrid = expand.grid(kmax = 5, distance = 2, kernel = "optimal"))

pre_KNNk5 <- predict(KNNk5, newdata = process_test)

conmatrix_KNNk5 <- table(pre_KNNk5, changedtpye_gender_submission$Survived)

print(conmatrix_KNNk5)

```

```

##
## pre_KNNk5    0    1
##              0 223  42
##              1  43 110

```

```

TP_KNNk5 <- sum(pre_KNNk5 == 1 & changedtpye_gender_submission$Survived == 1)
TN_KNNk5 <- sum(pre_KNNk5 == 0 & changedtpye_gender_submission$Survived == 0)
FP_KNNk5 <- sum(pre_KNNk5 == 1 & changedtpye_gender_submission$Survived == 0)
FN_KNNk5 <- sum(pre_KNNk5 == 0 & changedtpye_gender_submission$Survived == 1)

cat("True Positives in Titanic KNN:", TP_KNNk5, "\n")

```

```
## True Positives in Titanic KNN: 110
```

```
cat("True Negatives in Titanic KNN:", TN_KNNk5, "\n")
```

```
## True Negatives in Titanic KNN: 223
```

```
cat("False Positives in Titanic KNN:", FP_KNNk5, "\n")
```

```
## False Positives in Titanic KNN: 43
```

```
cat("False Negatives in Titanic KNN:", FN_KNNk5, "\n")
```

```
## False Negatives in Titanic KNN: 42
```

```

acc_KNNk5 <- (TP_KNNk5 + TN_KNNk5) / (TP_KNNk5 + TN_KNNk5 + FP_KNNk5 + FN_KNNk5)

precision_KNNk5 <- (TP_KNNk5) / (TP_KNNk5 + FP_KNNk5)

reca_KNNk5 <- (TP_KNNk5) / (TP_KNNk5 + FN_KNNk5)

```

```
f1_KNNk5 <- 2 * ((precision_KNNk5 * reca_KNNk5) / (precision_KNNk5 + reca_KNNk5))

cat("Accuracy of Titanic KNN:", acc_KNNk5, "\n")
```

```
## Accuracy of Titanic KNN: 0.7966507
```

```
cat("Precision of Titanic KNN:", precision_KNNk5, "\n")
```

```
## Precision of Titanic KNN: 0.7189542
```

```
cat("Recall of Titanic KNN:", reca_KNNk5, "\n")
```

```
## Recall of Titanic KNN: 0.7236842
```

```
cat("F1-Score of Titanic KNN:", f1_KNNk5, "\n")
```

```
## F1-Score of Titanic KNN: 0.7213115
```

KNN with k-fold cross-validation (k=4)

```
library(kknn)

process_test$Sex <- factor(process_test$Sex, levels = levels(process_train$Sex))
process_test$Nickname <- factor(process_test$Nickname, levels = levels(process_train$Nickname))

set.seed(1)
k4 <- trainControl(method = "cv", number = 4)

KNNk4 <- train(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,
               data = process_train, trControl = k4, method = "kknn",
               tuneGrid = expand.grid(kmax = 5, distance = 2, kernel = "optimal"))

pre_KNNk4 <- predict(KNNk4, newdata = process_test)

conmatrix_KNNk4 <- table(pre_KNNk4, changedtpye_gender_submission$Survived)

print(conmatrix_KNNk4)
```

```
##
## pre_KNNk4    0    1
##           0 223  42
##           1  43 110
```

```

TP_KNNk4 <- sum(pre_KNNk4 == 1 & changedtpye_gender_submission$Survived == 1)
TN_KNNk4 <- sum(pre_KNNk4 == 0 & changedtpye_gender_submission$Survived == 0)
FP_KNNk4 <- sum(pre_KNNk4 == 1 & changedtpye_gender_submission$Survived == 0)
FN_KNNk4 <- sum(pre_KNNk4 == 0 & changedtpye_gender_submission$Survived == 1)

cat("True Positives in Titanic KNN:", TP_KNNk4, "\n")

## True Positives in Titanic KNN: 110

cat("True Negatives in Titanic KNN:", TN_KNNk4, "\n")

## True Negatives in Titanic KNN: 223

cat("False Positives in Titanic KNN:", FP_KNNk4, "\n")

## False Positives in Titanic KNN: 43

cat("False Negatives in Titanic KNN:", FN_KNNk4, "\n")

## False Negatives in Titanic KNN: 42

acc_KNNk4 <- (TP_KNNk4 + TN_KNNk4) / (TP_KNNk4 + TN_KNNk4 + FP_KNNk4 + FN_KNNk4)
precision_KNNk4 <- (TP_KNNk4) / (TP_KNNk4 + FP_KNNk4)
reca_KNNk4 <- (TP_KNNk4) / (TP_KNNk4 + FN_KNNk4)
f1_KNNk4 <- 2 * ((precision_KNNk4 * reca_KNNk4) / (precision_KNNk4 + reca_KNNk4))

cat("Accuracy of Titanic KNN:", acc_KNNk4, "\n")

## Accuracy of Titanic KNN: 0.7966507

cat("Precision of Titanic KNN:", precision_KNNk4, "\n")

## Precision of Titanic KNN: 0.7189542

cat("Recall of Titanic KNN:", reca_KNNk4, "\n")

## Recall of Titanic KNN: 0.7236842

cat("F1-Score of Titanic KNN:", f1_KNNk4, "\n")

## F1-Score of Titanic KNN: 0.7213115

```

Support Vector Machine(SVM)

SVM with out cross-validation

```
library(e1071)

process_test$Sex <- factor(process_test$Sex, levels = levels(process_train$Sex))
process_test$Nickname <- factor(process_test$Nickname, levels = levels(process_train$Nickname))

set.seed(1)
SVMwithoutcross <- svm(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,
                        data = process_train,
                        kernel = "radial")

pred_SVMwithoutcross <- predict(SVMwithoutcross, newdata = process_test)

conmatrix_SVMwithoutcross <- table(pred_SVMwithoutcross, changedtpye_gender_submission$Survived)

print(conmatrix_SVMwithoutcross)

##
## pred_SVMwithoutcross    0    1
##                        0 250    6
##                        1   16 146

TP_SVMwithoutcross <- sum(pred_SVMwithoutcross == 1 & changedtpye_gender_submission$Survived == 1)
TN_SVMwithoutcross <- sum(pred_SVMwithoutcross == 0 & changedtpye_gender_submission$Survived == 0)
FP_SVMwithoutcross <- sum(pred_SVMwithoutcross == 1 & changedtpye_gender_submission$Survived == 0)
FN_SVMwithoutcross <- sum(pred_SVMwithoutcross == 0 & changedtpye_gender_submission$Survived == 1)

cat("True Positives in Titanic SVM:", TP_SVMwithoutcross, "\n")

## True Positives in Titanic SVM: 146

cat("True Negatives in Titanic SVM:", TN_SVMwithoutcross, "\n")

## True Negatives in Titanic SVM: 250

cat("False Positives in Titanic SVM:", FP_SVMwithoutcross, "\n")

## False Positives in Titanic SVM: 16
```

```
cat("False Negatives in Titanic SVM:", FN_SVMwithoutcross, "\n")
```

```
## False Negatives in Titanic SVM: 6
```

```
acc_SVMwithoutcross <- (TP_SVMwithoutcross + TN_SVMwithoutcross) / (TP_SVMwithoutcross + TN_SVMwithoutcross)
```

```
precision_SVMwithoutcross <- (TP_SVMwithoutcross) / (TP_SVMwithoutcross + FP_SVMwithoutcross)
```

```
reca_SVMwithoutcross <- (TP_SVMwithoutcross) / (TP_SVMwithoutcross + FN_SVMwithoutcross)
```

```
f1_SVMwithoutcross <- 2 * ((precision_SVMwithoutcross * reca_SVMwithoutcross) / (precision_SVMwithoutcross + reca_SVMwithoutcross))
```

```
cat("Accuracy of Titanic SVM:", acc_SVMwithoutcross, "\n")
```

```
## Accuracy of Titanic SVM: 0.9473684
```

```
cat("Precision of Titanic SVM:", precision_SVMwithoutcross, "\n")
```

```
## Precision of Titanic SVM: 0.9012346
```

```
cat("Recall of Titanic SVM:", reca_SVMwithoutcross, "\n")
```

```
## Recall of Titanic SVM: 0.9605263
```

```
cat("F1-Score of Titanic SVM:", f1_SVMwithoutcross, "\n")
```

```
## F1-Score of Titanic SVM: 0.9299363
```

SVM with k-fold cross-validation (k=10)

```
library(e1071)
```

```
library(caret)
```

```
process_test$Sex <- factor(process_test$Sex, levels = levels(process_train$Sex))
```

```
process_test$Nickname <- factor(process_test$Nickname, levels = levels(process_train$Nickname))
```

```
set.seed(1)
```

```
k10 <- trainControl(method = "cv", number = 10)
```

```
SVMk10 <- train(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,  
               data = process_train,  
               method = "svmRadial",  
               trControl = k10,  
               tuneLength = 10)
```

```
## Warning in .local(x, ...): Variable(s) ‘ ’ constant. Cannot scale data.
```

```
## Warning in .local(x, ...): Variable(s) ‘ ’ constant. Cannot scale data.
```

[illegible]

```
## Warning in .local(x, ...): Variable(s) ' ' constant. Cannot scale data.
## Warning in .local(x, ...): Variable(s) ' ' constant. Cannot scale data.
## Warning in .local(x, ...): Variable(s) ' ' constant. Cannot scale data.
## Warning in .local(x, ...): Variable(s) ' ' constant. Cannot scale data.
```

```
pred_SVMk10 <- predict(SVMk10, newdata = process_test)

conmatrix_SVMk10 <- table(pred_SVMk10, changedtpye_gender_submission$Survived)
print(conmatrix_SVMk10)
```

```
##
## pred_SVMk10    0    1
##              0 247   10
##              1   19  142
```

```
TP_SVMk10 <- sum(pred_SVMk10 == 1 & changedtpye_gender_submission$Survived == 1)

TN_SVMk10 <- sum(pred_SVMk10 == 0 & changedtpye_gender_submission$Survived == 0)

FP_SVMk10 <- sum(pred_SVMk10 == 1 & changedtpye_gender_submission$Survived == 0)

FN_SVMk10 <- sum(pred_SVMk10 == 0 & changedtpye_gender_submission$Survived == 1)

cat("True Positives in Titanic SVM:", TP_SVMk10, "\n")
```

```
## True Positives in Titanic SVM: 142
```

```
cat("True Negatives in Titanic SVM:", TN_SVMk10, "\n")
```

```
## True Negatives in Titanic SVM: 247
```

```
cat("False Positives in Titanic SVM:", FP_SVMk10, "\n")
```

```
## False Positives in Titanic SVM: 19
```

```
cat("False Negatives in Titanic SVM:", FN_SVMk10, "\n")
```

```
## False Negatives in Titanic SVM: 10
```

```
acc_SVMk10 <- (TP_SVMk10 + TN_SVMk10) / (TP_SVMk10 + TN_SVMk10 + FP_SVMk10 + FN_SVMk10)

precision_SVMk10 <- (TP_SVMk10) / (TP_SVMk10 + FP_SVMk10)

reca_SVMk10 <- (TP_SVMk10) / (TP_SVMk10 + FN_SVMk10)

f1_SVMk10 <- 2 * ((precision_SVMk10 * reca_SVMk10) / (precision_SVMk10 + reca_SVMk10))

cat("Accuracy of Titanic SVM:", acc_SVMk10, "\n")
```

```
## Accuracy of Titanic SVM: 0.930622
```

```
cat("Precision of Titanic SVM:", precision_SVMk10, "\n")
```

```
## Precision of Titanic SVM: 0.8819876
```

```
cat("Recall of Titanic SVM:", reca_SVMk10, "\n")
```

```
## Recall of Titanic SVM: 0.9342105
```

```
cat("F1-Score of Titanic SVM:", f1_SVMk10, "\n")
```

```
## F1-Score of Titanic SVM: 0.9073482
```

SVM with k-fold cross-validation (k=5)

```
library(e1071)
library(caret)

process_test$Sex <- factor(process_test$Sex, levels = levels(process_train$Sex))
process_test$Nickname <- factor(process_test$Nickname, levels = levels(process_train$Nickname))

set.seed(1)
k5 <- trainControl(method = "cv", number = 5)
SVMk5 <- train(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,
               data = process_train,
               method = "svmRadial",
               trControl = k5,
               tuneLength = 10)
```

```
## Warning in .local(x, ...): Variable(s) ' constant. Cannot scale data.
## Warning in .local(x, ...): Variable(s) ' constant. Cannot scale data.
## Warning in .local(x, ...): Variable(s) ' constant. Cannot scale data.
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## Warning in .local(x, ...): Variable(s) ' constant. Cannot scale data.
## Warning in .local(x, ...): Variable(s) ' constant. Cannot scale data.
## Warning in .local(x, ...): Variable(s) ' constant. Cannot scale data.
## Warning in .local(x, ...): Variable(s) ' constant. Cannot scale data.
## Warning in .local(x, ...): Variable(s) ' constant. Cannot scale data.
## Warning in .local(x, ...): Variable(s) ' constant. Cannot scale data.
## Warning in .local(x, ...): Variable(s) ' constant. Cannot scale data.
## Warning in .local(x, ...): Variable(s) ' constant. Cannot scale data.
## Warning in .local(x, ...): Variable(s) ' constant. Cannot scale data.
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## Warning in .local(x, ...): Variable(s) ' constant. Cannot scale data.
## Warning in .local(x, ...): Variable(s) ' constant. Cannot scale data.
## Warning in .local(x, ...): Variable(s) ' constant. Cannot scale data.
```

```
## Warning in .local(x, ...): Variable(s) ' constant. Cannot scale data.
## Warning in .local(x, ...): Variable(s) ' constant. Cannot scale data.
## Warning in .local(x, ...): Variable(s) ' constant. Cannot scale data.
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## Warning in .local(x, ...): Variable(s) ' constant. Cannot scale data.
## Warning in .local(x, ...): Variable(s) ' constant. Cannot scale data.
## Warning in .local(x, ...): Variable(s) ' constant. Cannot scale data.
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## Warning in .local(x, ...): Variable(s) ' constant. Cannot scale data.
## Warning in .local(x, ...): Variable(s) ' constant. Cannot scale data.
## Warning in .local(x, ...): Variable(s) ' constant. Cannot scale data.
```

```
pred_SVMk5 <- predict(SVMk5, newdata = process_test)

conmatrix_SVMk5 <- table(pred_SVMk5, changedtpye_gender_submission$Survived)
print(conmatrix_SVMk5)
```

```
##
## pred_SVMk5    0    1
##              0 247  10
##              1  19 142
```

```
TP_SVMk5 <- sum(pred_SVMk5 == 1 & changedtpye_gender_submission$Survived == 1)

TN_SVMk5 <- sum(pred_SVMk5 == 0 & changedtpye_gender_submission$Survived == 0)

FP_SVMk5 <- sum(pred_SVMk5 == 1 & changedtpye_gender_submission$Survived == 0)

FN_SVMk5 <- sum(pred_SVMk5 == 0 & changedtpye_gender_submission$Survived == 1)

cat("True Positives in Titanic SVM:", TP_SVMk5, "\n")
```

```
## True Positives in Titanic SVM: 142
```

```
cat("True Negatives in Titanic SVM:", TN_SVMk5, "\n")
```

```
## True Negatives in Titanic SVM: 247
```

```
cat("False Positives in Titanic SVM:", FP_SVMk5, "\n")
```

```
## False Positives in Titanic SVM: 19
```

```
cat("False Negatives in Titanic SVM:", FN_SVMk5, "\n")
```

```
## False Negatives in Titanic SVM: 10
```

```
acc_SVMk5 <- (TP_SVMk5 + TN_SVMk5) / (TP_SVMk5 + TN_SVMk5 + FP_SVMk5 + FN_SVMk5)
```

```
precision_SVMk5 <- (TP_SVMk5) / (TP_SVMk5 + FP_SVMk5)
```

```
reca_SVMk5 <- (TP_SVMk5) / (TP_SVMk5 + FN_SVMk5)
```

```
f1_SVMk5 <- 2 * ((precision_SVMk5 * reca_SVMk5) / (precision_SVMk5 + reca_SVMk5))
```

```
cat("Accuracy of Titanic SVM", acc_SVMk5, "\n")
```

```
## Accuracy of Titanic SVM 0.930622
```

```
cat("Precision of Titanic SVM:", precision_SVMk5, "\n")
```

```
## Precision of Titanic SVM: 0.8819876
```

```
cat("Recall of Titanic SVM:", reca_SVMk5, "\n")
```

```
## Recall of Titanic SVM: 0.9342105
```

```
cat("F1-Score of Titanic SVM:", f1_SVMk5, "\n")
```

```
## F1-Score of Titanic SVM: 0.9073482
```

SVM with k-fold cross-validation (k=4)

```
library(e1071)
```

```
library(caret)
```

```
process_test$Sex <- factor(process_test$Sex, levels = levels(process_train$Sex))
```

```
process_test$Nickname <- factor(process_test$Nickname, levels = levels(process_train$Nickname))
```

```
set.seed(1)
```

```
k4 <- trainControl(method = "cv", number = 4)
```

```
SVMk4 <- train(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,  
              data = process_train,  
              method = "svmRadial",  
              trControl = k4,  
              tuneLength = 10)
```

```
## Warning in .local(x, ...): Variable(s) ' ' constant. Cannot scale data.
```

```
## Warning in .local(x, ...): Variable(s) ' ' constant. Cannot scale data.
```

```
pred_SVMk4 <- predict(SVMk4, newdata = process_test)

conmatrix_SVMk4 <- table(pred_SVMk4, changedtpye_gender_submission$Survived)
print(conmatrix_SVMk4)
```

```
TP_SVMk4 <- sum(pred_SVMk4 == 1 & changedtpye_gender_submission$Survived == 1)
TN_SVMk4 <- sum(pred_SVMk4 == 0 & changedtpye_gender_submission$Survived == 0)
FP_SVMk4 <- sum(pred_SVMk4 == 1 & changedtpye_gender_submission$Survived == 0)
FN_SVMk4 <- sum(pred_SVMk4 == 0 & changedtpye_gender_submission$Survived == 1)
cat("True Positives in Titanic SVM:", TP_SVMk4, "\n")
```

28

```
cat("True Negatives in Titanic SVM:", TN_SVMk4, "\n")
```

```
## True Negatives in Titanic SVM: 248
```

```
cat("False Positives in Titanic SVM:", FP_SVMk4, "\n")
```

```
## False Positives in Titanic SVM: 18
```

```
cat("False Negatives in Titanic SVM:", FN_SVMk4, "\n")
```

```
## False Negatives in Titanic SVM: 7
```

```
acc_SVMk4 <- (TP_SVMk4 + TN_SVMk4) / (TP_SVMk4 + TN_SVMk4 + FP_SVMk4 + FN_SVMk4)
```

```
precision_SVMk4 <- (TP_SVMk4) / (TP_SVMk4 + FP_SVMk4)
```

```
reca_SVMk4 <- (TP_SVMk4) / (TP_SVMk4 + FN_SVMk4)
```

```
f1_SVMk4 <- 2 * ((precision_SVMk4 * reca_SVMk4) / (precision_SVMk4 + reca_SVMk4))
```

```
cat("Accuracy of Titanic SVM:", acc_SVMk4, "\n")
```

```
## Accuracy of Titanic SVM: 0.9401914
```

```
cat("Precision of Titanic SVM:", precision_SVMk4, "\n")
```

```
## Precision of Titanic SVM: 0.8895706
```

```
cat("Recall of Titanic SVM:", reca_SVMk4, "\n")
```

```
## Recall of Titanic SVM: 0.9539474
```

```
cat("F1-Score of Titanic SVM:", f1_SVMk4, "\n")
```

```
## F1-Score of Titanic SVM: 0.9206349
```

Naive Bayes(NB)

NB with out cross-validation

```
library(e1071)
```

```
process_test$Sex <- factor(process_test$Sex, levels = levels(process_train$Sex))
```

```
process_test$Nickname <- factor(process_test$Nickname, levels = levels(process_train$Nickname))
```

```

set.seed(1)

naibawithoutcross <- naiveBayes(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,
                                data = process_train)

pred_naibawithoutcross <- predict(naibawithoutcross, newdata = process_test)

conmatrix_naibawithoutcross <- table(pred_naibawithoutcross, changedtpye_gender_submission$Survived)

print(conmatrix_naibawithoutcross)

##
## pred_naibawithoutcross    0    1
##                          0 260    5
##                          1    6 147

TP_naibawithoutcross <- sum(pred_naibawithoutcross == 1 & changedtpye_gender_submission$Survived == 1)
TN_naibawithoutcross <- sum(pred_naibawithoutcross == 0 & changedtpye_gender_submission$Survived == 0)
FP_naibawithoutcross <- sum(pred_naibawithoutcross == 1 & changedtpye_gender_submission$Survived == 0)
FN_naibawithoutcross <- sum(pred_naibawithoutcross == 0 & changedtpye_gender_submission$Survived == 1)

cat("True Positives in Titanic NB:", TP_naibawithoutcross, "\n")

## True Positives in Titanic NB: 147

cat("True Negatives in Titanic NB:", TN_naibawithoutcross, "\n")

## True Negatives in Titanic NB: 260

cat("False Positives in Titanic NB:", FP_naibawithoutcross, "\n")

## False Positives in Titanic NB: 6

cat("False Negatives in Titanic NB:", FN_naibawithoutcross, "\n")

## False Negatives in Titanic NB: 5

acc_naibawithoutcross <- (TP_naibawithoutcross + TN_naibawithoutcross) / (TP_naibawithoutcross + TN_naibawithoutcross + FP_naibawithoutcross + FN_naibawithoutcross)
precision_naibawithoutcross <- (TP_naibawithoutcross) / (TP_naibawithoutcross + FP_naibawithoutcross)
reca_naibawithoutcross <- (TP_naibawithoutcross) / (TP_naibawithoutcross + FN_naibawithoutcross)
f1_naibawithoutcross <- 2 * ((precision_naibawithoutcross * reca_naibawithoutcross) / (precision_naibawithoutcross + reca_naibawithoutcross))

cat("Accuracy of Titanic NB:", acc_naibawithoutcross, "\n")

```

```
## Accuracy of Titanic NB: 0.9736842
```

```
cat("Precision of Titanic NB:", precision_naibawithoutcross, "\n")
```

```
## Precision of Titanic NB: 0.9607843
```

```
cat("Recall of Titanic NB:", reca_naibawithoutcross, "\n")
```

```
## Recall of Titanic NB: 0.9671053
```

```
cat("F1-Score of Titanic NB:", f1_naibawithoutcross, "\n")
```

```
## F1-Score of Titanic NB: 0.9639344
```

NB with k-fold cross-validation (k=10)

```
unique(process_train$Nickname)
```

```
## [1] Mr
## [2] Mrs
## [3] Miss
## [4] Master
## [5] Don
## [6] Rev
## [7] Dr
## [8] Mme
## [9] Ms
## [10] Major
## [11] Lady
## [12] Sir
## [13] Mlle
## [14] Col
## [15] Capt
## [16] Rothes, the Countess. of (Lucy Noel Martha Dyer-Edwards)
## [17] Jonkheer
## 17 Levels: Capt Col Don Dr Jonkheer Lady Major Master Miss Mlle Mme Mr ... Sir
```

```
unique(process_test$Nickname)
```

```
## [1] Mr      Mrs      Miss     Master Ms      Col      Rev      Dr
## 17 Levels: Capt Col Don Dr Jonkheer Lady Major Master Miss Mlle Mme Mr ... Sir
```

```
unique(changedtpye_gender_submission$Nickname)
```

```
## NULL
```

```

library(caret)
library(dplyr)
library(forcats)

origprocess_train <- process_train
origprocess_test <- process_test

#Groups all Nickname levels that appear fewer than 5 times into an (Other) category
temprocess_train <- process_train %>%
  mutate(Nickname = fct_lump_min(Nickname, min = 5))

temprocess_test <- process_test %>%
  mutate(Nickname = fct_lump_min(Nickname, min = 5)) %>%
  mutate(Nickname = factor(Nickname, levels = levels(temprocess_train$Nickname)))

set.seed(1)
k10 <- trainControl(method = "cv", number = 10)

naibak10 <- train(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,
  data = temprocess_train,
  method = "nb",
  trControl = k10)

```

```

## Warning: model fit failed for Fold01: usekernel=FALSE, fL=0, adjust=1 Error in NaiveBayes.default(x,
##   Zero variances for at least one class in variables: NicknameRev

```

```

## Warning: model fit failed for Fold02: usekernel=FALSE, fL=0, adjust=1 Error in NaiveBayes.default(x,
##   Zero variances for at least one class in variables: NicknameRev

```

```

## Warning: model fit failed for Fold03: usekernel=FALSE, fL=0, adjust=1 Error in NaiveBayes.default(x,
##   Zero variances for at least one class in variables: NicknameRev

```

```

## Warning: model fit failed for Fold04: usekernel=FALSE, fL=0, adjust=1 Error in NaiveBayes.default(x,
##   Zero variances for at least one class in variables: NicknameRev

```

```

## Warning: model fit failed for Fold05: usekernel=FALSE, fL=0, adjust=1 Error in NaiveBayes.default(x,
##   Zero variances for at least one class in variables: NicknameRev

```

```

## Warning: model fit failed for Fold06: usekernel=FALSE, fL=0, adjust=1 Error in NaiveBayes.default(x,
##   Zero variances for at least one class in variables: NicknameRev

```

```

## Warning: model fit failed for Fold07: usekernel=FALSE, fL=0, adjust=1 Error in NaiveBayes.default(x,
##   Zero variances for at least one class in variables: NicknameRev

```

```

## Warning: model fit failed for Fold08: usekernel=FALSE, fL=0, adjust=1 Error in NaiveBayes.default(x,
##   Zero variances for at least one class in variables: NicknameRev

```

```

## Warning: model fit failed for Fold09: usekernel=FALSE, fL=0, adjust=1 Error in NaiveBayes.default(x,
##   Zero variances for at least one class in variables: NicknameRev

```

```
## Warning: model fit failed for Fold10: usekernel=FALSE, fL=0, adjust=1 Error in NaiveBayes.default(x,
##   Zero variances for at least one class in variables: NicknameRev
```

```
## Warning in nominalTrainWorkflow(x = x, y = y, wts = weights, info = trainInfo,
## : There were missing values in resampled performance measures.
```

```
## Warning in train.default(x, y, weights = w, ...): missing values found in
## aggregated results
```

```
pred_naibak10 <- predict(naibak10, newdata = temprocess_test)

conmatrix_naibak10 <- table(pred_naibak10, changedtpye_gender_submission$Survived)
print(conmatrix_naibak10)
```

```
##
## pred_naibak10    0    1
##                0 266 142
##                1    0  10
```

```
process_train <- origprocess_train
process_test  <- origprocess_test
```

```
TP_naibak10 <- sum(pred_naibak10 == 1 & changedtpye_gender_submission$Survived == 1)
TN_naibak10 <- sum(pred_naibak10 == 0 & changedtpye_gender_submission$Survived == 0)
FP_naibak10 <- sum(pred_naibak10 == 1 & changedtpye_gender_submission$Survived == 0)
FN_naibak10 <- sum(pred_naibak10 == 0 & changedtpye_gender_submission$Survived == 1)

cat("True Positives in Titanic NB:", TP_naibak10, "\n")
```

```
## True Positives in Titanic NB: 10
```

```
cat("True Negatives in Titanic NB:", TN_naibak10, "\n")
```

```
## True Negatives in Titanic NB: 266
```

```
cat("False Positives in Titanic NB:", FP_naibak10, "\n")
```

```
## False Positives in Titanic NB: 0
```

```
cat("False Negatives in Titanic NB:", FN_naibak10, "\n")
```

```
## False Negatives in Titanic NB: 142
```

```
table(pred_naibak10, changedtpye_gender_submission$Survived)
```

```
##
## pred_naibak10    0    1
##                0 266 142
##                1    0   10

acc_naibak10 <- (TP_naibak10 + TN_naibak10) / (TP_naibak10 + TN_naibak10 + FP_naibak10 + FN_naibak10)

precision_naibak10 <- (TP_naibak10) / (TP_naibak10 + FP_naibak10)

reca_naibak10 <- (TP_naibak10) / (TP_naibak10 + FN_naibak10)

f1_naibak10 <- 2 * ((precision_naibak10 * reca_naibak10) / (precision_naibak10 + reca_naibak10))

cat("Accuracy of Titanic NB:", acc_naibak10, "\n")

## Accuracy of Titanic NB: 0.6602871

cat("Precision of Titanic NB:", precision_naibak10, "\n")

## Precision of Titanic NB: 1

cat("Recall of Titanic NB:", reca_naibak10, "\n")

## Recall of Titanic NB: 0.06578947

cat("F1-Score of Titanic NB:", f1_naibak10, "\n")

## F1-Score of Titanic NB: 0.1234568
```

NB with k-fold cross-validation (k=5)

```
library(caret)
library(dplyr)
library(forcats)

origprocess_train <- process_train
origprocess_test <- process_test

#Groups all Nickname levels that appear fewer than 5 times into an (Other) category
temprocess_train <- process_train %>%
  mutate(Nickname = fct_lump_min(Nickname, min = 5))

temprocess_test <- process_test %>%
  mutate(Nickname = fct_lump_min(Nickname, min = 5)) %>%
  mutate(Nickname = factor(Nickname, levels = levels(temprocess_train$Nickname)))

set.seed(1)
k5 <- trainControl(method = "cv", number = 5)
```

```
naibak5 <- train(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,
  data = temprocess_train,
  method = "nb",
  trControl = k5)
```

```
## Warning: model fit failed for Fold1: usekernel=FALSE, fL=0, adjust=1 Error in NaiveBayes.default(x, y)
## Zero variances for at least one class in variables: NicknameRev
```

```
## Warning: model fit failed for Fold2: usekernel=FALSE, fL=0, adjust=1 Error in NaiveBayes.default(x, y)
## Zero variances for at least one class in variables: NicknameRev
```

```
## Warning: model fit failed for Fold3: usekernel=FALSE, fL=0, adjust=1 Error in NaiveBayes.default(x, y)
## Zero variances for at least one class in variables: NicknameRev
```

```
## Warning: model fit failed for Fold4: usekernel=FALSE, fL=0, adjust=1 Error in NaiveBayes.default(x, y)
## Zero variances for at least one class in variables: NicknameRev
```

```
## Warning: model fit failed for Fold5: usekernel=FALSE, fL=0, adjust=1 Error in NaiveBayes.default(x, y)
## Zero variances for at least one class in variables: NicknameRev
```

```
## Warning in nominalTrainWorkflow(x = x, y = y, wts = weights, info = trainInfo,
## : There were missing values in resampled performance measures.
```

```
## Warning in train.default(x, y, weights = w, ...): missing values found in
## aggregated results
```

```
pred_naibak5 <- predict(naibak5, newdata = temprocess_test)
```

```
conmatrix_naibak5 <- table(pred_naibak5, changedtpye_gender_submission$Survived)
print(conmatrix_naibak5)
```

```
##
## pred_naibak5    0    1
##               0 266 142
##               1    0   10
```

```
process_train <- origprocess_train
process_test <- origprocess_test
```

```
TP_naibak5 <- sum(pred_naibak5 == 1 & changedtpye_gender_submission$Survived == 1)
```

```
TN_naibak5 <- sum(pred_naibak5 == 0 & changedtpye_gender_submission$Survived == 0)
```

```
FP_naibak5 <- sum(pred_naibak5 == 1 & changedtpye_gender_submission$Survived == 0)
```

```
FN_naibak5 <- sum(pred_naibak5 == 0 & changedtpye_gender_submission$Survived == 1)
```

```
cat("True Positives in Titanic NB:", TP_naibak5, "\n")
```

```
## True Positives in Titanic NB: 10
```

```

cat("True Negatives in Titanic NB:", TN_naibak5, "\n")

## True Negatives in Titanic NB: 266

cat("False Positives in Titanic NB:", FP_naibak5, "\n")

## False Positives in Titanic NB: 0

cat("False Negatives in Titanic NB:", FN_naibak5, "\n")

## False Negatives in Titanic NB: 142

acc_naibak5 <- (TP_naibak5 + TN_naibak5) / (TP_naibak5 + TN_naibak5 + FP_naibak5 + FN_naibak5)
precision_naibak5 <- (TP_naibak5) / (TP_naibak5 + FP_naibak5)
reca_naibak5 <- (TP_naibak5) / (TP_naibak5 + FN_naibak5)
f1_naibak5 <- 2 * ((precision_naibak5 * reca_naibak5) / (precision_naibak5 + reca_naibak5))
cat("Accuracy of Titanic NB:", acc_naibak5, "\n")

## Accuracy of Titanic NB: 0.6602871

cat("Precision of Titanic NB:", precision_naibak5, "\n")

## Precision of Titanic NB: 1

cat("Recall of Titanic NB:", reca_naibak5, "\n")

## Recall of Titanic NB: 0.06578947

cat("F1-Score of Titanic NB:", f1_naibak5, "\n")

## F1-Score of Titanic NB: 0.1234568

```

NB with k-fold cross-validation (k=4)

```

library(caret)
library(dplyr)
library(forcats)

origprocess_train <- process_train
origprocess_test <- process_test

```

```

#Groups all Nickname levels that appear fewer than 5 times into an (Other) category
temprocess_train <- process_train %>%
  mutate(Nickname = fct_lump_min(Nickname, min = 5))

temprocess_test <- process_test %>%
  mutate(Nickname = fct_lump_min(Nickname, min = 5)) %>%
  mutate(Nickname = factor(Nickname, levels = levels(temprocess_train$Nickname)))

set.seed(1)
k4 <- trainControl(method = "cv", number = 4)

naibak4 <- train(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,
  data = temprocess_train,
  method = "nb",
  trControl = k4)

```

```

## Warning: model fit failed for Fold1: usekernel=FALSE, fL=0, adjust=1 Error in NaiveBayes.default(x, y,
##   Zero variances for at least one class in variables: NicknameRev

```

```

## Warning: model fit failed for Fold2: usekernel=FALSE, fL=0, adjust=1 Error in NaiveBayes.default(x, y,
##   Zero variances for at least one class in variables: NicknameRev

```

```

## Warning: model fit failed for Fold3: usekernel=FALSE, fL=0, adjust=1 Error in NaiveBayes.default(x, y,
##   Zero variances for at least one class in variables: NicknameRev

```

```

## Warning: model fit failed for Fold4: usekernel=FALSE, fL=0, adjust=1 Error in NaiveBayes.default(x, y,
##   Zero variances for at least one class in variables: NicknameRev

```

```

## Warning in nominalTrainWorkflow(x = x, y = y, wts = weights, info = trainInfo,
## : There were missing values in resampled performance measures.

```

```

## Warning in train.default(x, y, weights = w, ...): missing values found in
## aggregated results

```

```

pred_naibak4 <- predict(naibak4, newdata = temprocess_test)

conmatrix_naibak4 <- table(pred_naibak4, changedtpye_gender_submission$Survived)
print(conmatrix_naibak4)

```

```

##
## pred_naibak4    0    1
##               0 266 142
##               1   0  10

```

```

process_train <- origprocess_train
process_test <- origprocess_test

```

```

TP_naibak4 <- sum(pred_naibak4 == 1 & changedtpye_gender_submission$Survived == 1)

```

```

TN_naibak4 <- sum(pred_naibak4 == 0 & changedtpye_gender_submission$Survived == 0)

```

```

FP_naibak4 <- sum(pred_naibak4 == 1 & changedtpye_gender_submission$Survived == 0)
FN_naibak4 <- sum(pred_naibak4 == 0 & changedtpye_gender_submission$Survived == 1)

cat("True Positives in Titanic NB:", TP_naibak4, "\n")

## True Positives in Titanic NB: 10

cat("True Negatives in Titanic NB:", TN_naibak4, "\n")

## True Negatives in Titanic NB: 266

cat("False Positives in Titanic NB:", FP_naibak4, "\n")

## False Positives in Titanic NB: 0

cat("False Negatives in Titanic NB:", FN_naibak4, "\n")

## False Negatives in Titanic NB: 142

table(pred_naibak4, changedtpye_gender_submission$Survived)

##
## pred_naibak4    0    1
##              0 266 142
##              1   0  10

acc_naibak4 <- (TP_naibak4 + TN_naibak4) / (TP_naibak4 + TN_naibak4 + FP_naibak4 + FN_naibak4)
precision_naibak4 <- (TP_naibak4) / (TP_naibak4 + FP_naibak4)
reca_naibak4 <- (TP_naibak4) / (TP_naibak4 + FN_naibak4)
f1_naibak4 <- 2 * ((precision_naibak4 * reca_naibak4) / (precision_naibak4 + reca_naibak4))

cat("Accuracy of Titanic NB:", acc_naibak4, "\n")

## Accuracy of Titanic NB: 0.6602871

cat("Precision of Titanic NB:", precision_naibak4, "\n")

## Precision of Titanic NB: 1

cat("Recall of Titanic NB:", reca_naibak4, "\n")

## Recall of Titanic NB: 0.06578947

```

```
cat("F1-Score of Titanic NB:", f1_naibak4, "\n")
```

```
## F1-Score of Titanic NB: 0.1234568
```

Decision Tree (DT)

DT with out cross-validation

```
library(rpart)
library(caret)
library(dplyr)
library(forcats)

origprocess_train <- process_train
origprocess_test <- process_test

# Group all Nickname levels that appear fewer than 5 times into an (Other) category
temprocess_train <- process_train %>%
  mutate(Nickname = fct_lump_min(Nickname, min = 5))

temprocess_test <- process_test %>%
  mutate(Nickname = fct_lump_min(Nickname, min = 5)) %>%
  mutate(Nickname = factor(Nickname, levels = levels(temprocess_train$Nickname)))

set.seed(1)

treebestofdepth <- NULL
treebestofaccuracy <- 0

# try different depth of the tree
for(depth in c(5, 10, 20, 25, 30)) {
  set.seed(1)

  dectrwithoutcross <- rpart(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,
    data = temprocess_train,
    control = rpart.control(maxdepth = depth))

  print(paste("DT summary of max depth:", depth))

  pred_dectrwithoutcross <- predict(dectrwithoutcross, newdata = temprocess_test, type = "class")

  conmatrix_dectrwithoutcross <- table(pred_dectrwithoutcross, changedtpye_gender_submission$Survived)
  print(conmatrix_dectrwithoutcross)

  accuracy <- sum(diag(conmatrix_dectrwithoutcross)) / sum(conmatrix_dectrwithoutcross)

  if(accuracy > treebestofaccuracy) {
    treebestofaccuracy <- accuracy
    treebestofdepth <- depth
  }
}
```

```
}
}
```

```
## [1] "DT summary of max depth: 5"
##
## pred_dectrwithoutcross    0    1
##                0 245    6
##                1  21 146
## [1] "DT summary of max depth: 10"
##
## pred_dectrwithoutcross    0    1
##                0 245    6
##                1  21 146
## [1] "DT summary of max depth: 20"
##
## pred_dectrwithoutcross    0    1
##                0 245    6
##                1  21 146
## [1] "DT summary of max depth: 25"
##
## pred_dectrwithoutcross    0    1
##                0 245    6
##                1  21 146
## [1] "DT summary of max depth: 30"
##
## pred_dectrwithoutcross    0    1
##                0 245    6
##                1  21 146
```

```
process_train <- origprocess_train
process_test  <- origprocess_test
```

```
print(paste("Best Max Depth:", treebestofdepth, "with accuracy:", round(treebestofaccuracy * 100, 4), "%"))
```

```
## [1] "Best Max Depth: 5 with accuracy: 93.5407 %"
```

```
TP_dectrwithoutcross <- sum(pred_dectrwithoutcross == 1 & changedtpye_gender_submission$Survived == 1)
TN_dectrwithoutcross <- sum(pred_dectrwithoutcross == 0 & changedtpye_gender_submission$Survived == 0)
FP_dectrwithoutcross <- sum(pred_dectrwithoutcross == 1 & changedtpye_gender_submission$Survived == 0)
FN_dectrwithoutcross <- sum(pred_dectrwithoutcross == 0 & changedtpye_gender_submission$Survived == 1)
cat("True Positives in Titanic DT:", TP_dectrwithoutcross, "\n")
```

```
## True Positives in Titanic DT: 146
```

```
cat("True Negatives in Titanic DT:", TN_dectrwithoutcross, "\n")
```

```
## True Negatives in Titanic DT: 245
```

```

cat("False Positives in Titanic DT:", FP_dectrwithoutcross, "\n")

## False Positives in Titanic DT: 21

cat("False Negatives in Titanic DT:", FN_dectrwithoutcross, "\n")

## False Negatives in Titanic DT: 6

# Check lengths
length(pred_dectrwithoutcross)

## [1] 418

length(changedtpye_gender_submission$Survived)

## [1] 418

table(pred_dectrwithoutcross, changedtpye_gender_submission$Survived)

##
## pred_dectrwithoutcross    0    1
##                0 245    6
##                1  21 146

acc_dectrwithoutcross <- (TP_dectrwithoutcross + TN_dectrwithoutcross) / (TP_dectrwithoutcross + TN_dectrwithoutcross)
precision_dectrwithoutcross <- (TP_dectrwithoutcross) / (TP_dectrwithoutcross + FP_dectrwithoutcross)
reca_dectrwithoutcross <- (TP_dectrwithoutcross) / (TP_dectrwithoutcross + FN_dectrwithoutcross)
f1_dectrwithoutcross <- 2 * ((precision_dectrwithoutcross * reca_dectrwithoutcross) / (precision_dectrwithoutcross + reca_dectrwithoutcross))

cat("Accuracy of Titanic DT:", acc_dectrwithoutcross, "\n")

## Accuracy of Titanic DT: 0.9354067

cat("Precision of Titanic DT:", precision_dectrwithoutcross, "\n")

## Precision of Titanic DT: 0.8742515

cat("Recall of Titanic DT:", reca_dectrwithoutcross, "\n")

## Recall of Titanic DT: 0.9605263

cat("F1-Score of Titanic DT:", f1_dectrwithoutcross, "\n")

## F1-Score of Titanic DT: 0.9153605

```

DT with k-fold cross-validation (k=10)

```

library(rpart)
library(caret)
library(dplyr)
library(forcats)

origprocess_train <- process_train
origprocess_test <- process_test

# Group all Nickname levels that appear fewer than 5 times into an (Other) category
temprocess_train <- process_train %>%
  mutate(Nickname = fct_lump_min(Nickname, min = 5))

temprocess_test <- process_test %>%
  mutate(Nickname = fct_lump_min(Nickname, min = 5)) %>%
  mutate(Nickname = factor(Nickname, levels = levels(temprocess_train$Nickname)))

set.seed(1)

treebestofdepth <- NULL
treebestofaccuracy <- 0

k10 <- trainControl(method = "cv", number = 10)

# try different depth of the tree
sampledepth <- expand.grid(maxdepth = c(5, 10, 20, 25, 30))

dectrk10 <- train(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,
  data = temprocess_train,
  method = "rpart2",
  tuneGrid = sampledepth,
  trControl = k10)

print(paste("DT summary of max depth:", depth))

## [1] "DT summary of max depth: 30"

pred_dectrk10 <- predict(dectrk10, newdata = temprocess_test)

conmatrix_dectrk10 <- table(pred_dectrk10, changedtpye_gender_submission$Survived)
print(conmatrix_dectrk10)

##
## pred_dectrk10    0    1
##               0 250    6
##               1  16 146

treebestmodel <- dectrk10$bestTune

accuracy <- sum(diag(conmatrix_dectrk10)) / sum(conmatrix_dectrk10)

```

```

print(paste("Best Max Depth:", treebestmodel$maxdepth, "with accuracy:", round(accuracy * 100, 4), "%"))

## [1] "Best Max Depth: 5 with accuracy: 94.7368 %"

process_train <- origprocess_train
process_test <- origprocess_test

TP_dectrk10 <- sum(pred_dectrk10 == 1 & changedtpye_gender_submission$Survived == 1)
TN_dectrk10 <- sum(pred_dectrk10 == 0 & changedtpye_gender_submission$Survived == 0)
FP_dectrk10 <- sum(pred_dectrk10 == 1 & changedtpye_gender_submission$Survived == 0)
FN_dectrk10 <- sum(pred_dectrk10 == 0 & changedtpye_gender_submission$Survived == 1)

cat("True Positives in Titanic DT:", TP_dectrk10, "\n")

## True Positives in Titanic DT: 146

cat("True Negatives in Titanic DT:", TN_dectrk10, "\n")

## True Negatives in Titanic DT: 250

cat("False Positives in Titanic DT:", FP_dectrk10, "\n")

## False Positives in Titanic DT: 16

cat("False Negatives in Titanic DT:", FN_dectrk10, "\n")

## False Negatives in Titanic DT: 6

table(pred_dectrk10, changedtpye_gender_submission$Survived)

##
## pred_dectrk10    0    1
##                0 250    6
##                1  16 146

acc_dectrk10 <- (TP_dectrk10 + TN_dectrk10) / (TP_dectrk10 + TN_dectrk10 + FP_dectrk10 + FN_dectrk10)
precision_dectrk10 <- (TP_dectrk10) / (TP_dectrk10 + FP_dectrk10)
reca_dectrk10 <- (TP_dectrk10) / (TP_dectrk10 + FN_dectrk10)
f1_dectrk10 <- 2 * ((precision_dectrk10 * reca_dectrk10) / (precision_dectrk10 + reca_dectrk10))

cat("Accuracy of Titanic DT:", acc_dectrk10, "\n")

## Accuracy of Titanic DT: 0.9473684

```

```
cat("Precision of Titanic DT:", precision_dectrk10, "\n")
```

```
## Precision of Titanic DT: 0.9012346
```

```
cat("Recall of Titanic DT:", reca_dectrk10, "\n")
```

```
## Recall of Titanic DT: 0.9605263
```

```
cat("F1-Score of Titanic DT:", f1_dectrk10, "\n")
```

```
## F1-Score of Titanic DT: 0.9299363
```

DT with k-fold cross-validation (k=5)

```
library(rpart)
library(caret)
library(dplyr)
library(forcats)

origprocess_train <- process_train
origprocess_test  <- process_test

# Group all Nickname levels that appear fewer than 5 times into an (Other) category
temprocess_train <- process_train %>%
  mutate(Nickname = fct_lump_min(Nickname, min = 5))

temprocess_test <- process_test %>%
  mutate(Nickname = fct_lump_min(Nickname, min = 5)) %>%
  mutate(Nickname = factor(Nickname, levels = levels(temprocess_train$Nickname)))

set.seed(1)

treebestofdepth <- NULL
treebestofaccuracy <- 0

k5 <- trainControl(method = "cv", number = 5)

# try different depth of the tree
sampledepth <- expand.grid(maxdepth = c(5, 10, 20, 25, 30))

dectrk5 <- train(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,
  data = temprocess_train,
  method = "rpart2",
  tuneGrid = sampledepth,
  trControl = k5)

print(paste("DT summary of max depth:", depth))

## [1] "DT summary of max depth: 30"
```

```

pred_dectrk5 <- predict(detrk5, newdata = temprocess_test)

conmatrix_dectrk5 <- table(pred_dectrk5, changedtpye_gender_submission$Survived)
print(conmatrix_dectrk5)

##
## pred_dectrk5    0    1
##              0 250    6
##              1  16 146

treebestmodel <- detrk5$bestTune

accuracy <- sum(diag(conmatrix_dectrk5)) / sum(conmatrix_dectrk5)

print(paste("Best Max Depth:", treebestmodel$maxdepth, "with accuracy:", round(accuracy * 100, 4), "%"))

## [1] "Best Max Depth: 5 with accuracy: 94.7368 %"

process_train <- origprocess_train
process_test <- origprocess_test

TP_dectrk5 <- sum(pred_dectrk5 == 1 & changedtpye_gender_submission$Survived == 1)
TN_dectrk5 <- sum(pred_dectrk5 == 0 & changedtpye_gender_submission$Survived == 0)
FP_dectrk5 <- sum(pred_dectrk5 == 1 & changedtpye_gender_submission$Survived == 0)
FN_dectrk5 <- sum(pred_dectrk5 == 0 & changedtpye_gender_submission$Survived == 1)

cat("True Positives in Titanic DT:", TP_dectrk5, "\n")

## True Positives in Titanic DT: 146

cat("True Negatives in Titanic DT:", TN_dectrk5, "\n")

## True Negatives in Titanic DT: 250

cat("False Positives in Titanic DT:", FP_dectrk5, "\n")

## False Positives in Titanic DT: 16

cat("False Negatives in Titanic DT:", FN_dectrk5, "\n")

## False Negatives in Titanic DT: 6

table(pred_dectrk5, changedtpye_gender_submission$Survived)

```

```
##
## pred_dectrk5    0    1
##               0 250    6
##               1  16 146

acc_dectrk5 <- (TP_dectrk5 + TN_dectrk5) / (TP_dectrk5 + TN_dectrk5 + FP_dectrk5 + FN_dectrk5)

precision_dectrk5 <- (TP_dectrk5) / (TP_dectrk5 + FP_dectrk5)

reca_dectrk5 <- (TP_dectrk5) / (TP_dectrk5 + FN_dectrk5)

f1_dectrk5 <- 2 * ((precision_dectrk5 * reca_dectrk5) / (precision_dectrk5 + reca_dectrk5))

cat("Accuracy of Titanic DT:", acc_dectrk5, "\n")

## Accuracy of Titanic DT: 0.9473684

cat("Precision of Titanic DT:", precision_dectrk5, "\n")

## Precision of Titanic DT: 0.9012346

cat("Recall of Titanic DT:", reca_dectrk5, "\n")

## Recall of Titanic DT: 0.9605263

cat("F1-Score of Titanic DT:", f1_dectrk5, "\n")

## F1-Score of Titanic DT: 0.9299363
```

DT with k-fold cross-validation (k=4)

```
library(rpart)
library(caret)
library(dplyr)
library(forcats)

origprocess_train <- process_train
origprocess_test  <- process_test

# Group all Nickname levels that appear fewer than 5 times into an (Other) category
temprocess_train <- process_train %>%
  mutate(Nickname = fct_lump_min(Nickname, min = 5))

temprocess_test <- process_test %>%
  mutate(Nickname = fct_lump_min(Nickname, min = 5)) %>%
  mutate(Nickname = factor(Nickname, levels = levels(temprocess_train$Nickname)))

set.seed(1)
```

```

treebestofdepth <- NULL
treebestofaccuracy <- 0

k4 <- trainControl(method = "cv", number = 4)

# try different depth of the tree
sampledepth <- expand.grid(maxdepth = c(5, 10, 20, 25, 30))

dectrk4 <- train(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,
                 data = temprocess_train,
                 method = "rpart2",
                 tuneGrid = sampledepth,
                 trControl = k4)

print(paste("DT summary of max depth:", depth))

## [1] "DT summary of max depth: 30"

pred_dectrk4 <- predict(dectrk4, newdata = temprocess_test)

conmatrix_dectrk4 <- table(pred_dectrk4, changedtpye_gender_submission$Survived)
print(conmatrix_dectrk4)

##
## pred_dectrk4    0    1
##           0 250    6
##           1  16 146

treebestmodel <- dectrk4$bestTune

accuracy <- sum(diag(conmatrix_dectrk4)) / sum(conmatrix_dectrk4)

print(paste("Best Max Depth:", treebestmodel$maxdepth, "with accuracy:", round(accuracy * 100, 4), "%"))

## [1] "Best Max Depth: 5 with accuracy: 94.7368 %"

process_train <- origprocess_train
process_test <- origprocess_test

TP_dectrk4 <- sum(pred_dectrk4 == 1 & changedtpye_gender_submission$Survived == 1)
TN_dectrk4 <- sum(pred_dectrk4 == 0 & changedtpye_gender_submission$Survived == 0)
FP_dectrk4 <- sum(pred_dectrk4 == 1 & changedtpye_gender_submission$Survived == 0)
FN_dectrk4 <- sum(pred_dectrk4 == 0 & changedtpye_gender_submission$Survived == 1)

cat("True Positives in Titanic DT:", TP_dectrk4, "\n")

## True Positives in Titanic DT: 146

```

```

cat("True Negatives in Titanic DT:", TN_dectrk4, "\n")

## True Negatives in Titanic DT: 250

cat("False Positives in Titanic DT:", FP_dectrk4, "\n")

## False Positives in Titanic DT: 16

cat("False Negatives in Titanic DT:", FN_dectrk4, "\n")

## False Negatives in Titanic DT: 6

table(pred_dectrk4, changedtpye_gender_submission$Survived)

##
## pred_dectrk4    0    1
##              0 250    6
##              1  16 146

acc_dectrk4 <- (TP_dectrk4 + TN_dectrk4) / (TP_dectrk4 + TN_dectrk4 + FP_dectrk4 + FN_dectrk4)

precision_dectrk4 <- (TP_dectrk4) / (TP_dectrk4 + FP_dectrk4)

reca_dectrk4 <- (TP_dectrk4) / (TP_dectrk4 + FN_dectrk4)

f1_dectrk4 <- 2 * ((precision_dectrk4 * reca_dectrk4) / (precision_dectrk4 + reca_dectrk4))

cat("Accuracy of Titanic DT:", acc_dectrk4, "\n")

## Accuracy of Titanic DT: 0.9473684

cat("Precision of Titanic DT:", precision_dectrk4, "\n")

## Precision of Titanic DT: 0.9012346

cat("Recall of Titanic DT:", reca_dectrk4, "\n")

## Recall of Titanic DT: 0.9605263

cat("F1-Score of Titanic DT:", f1_dectrk4, "\n")

## F1-Score of Titanic DT: 0.9299363

```

Random Forest(RF)

RF with out cross-validation

```

library(randomForest)

process_test$Sex <- factor(process_test$Sex, levels = levels(process_train$Sex))
process_test$Nickname <- factor(process_test$Nickname, levels = levels(process_train$Nickname))

set.seed(1)

# find the best tree
samplenumtree <- c(10, 100, 250, 500, 1000)
samplemaxnodes <- c(10, 20, 30, 50, 100)

setbestnumtree <- 0
setbestmaxnodes <- 0
setbestaccuracy <- 0

for (ntree in samplenumtree) {
  for (maxnodes in samplemaxnodes) {
    ranfomodel <- randomForest(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,
                              data = process_train,
                              ntree = ntree,
                              maxnodes = maxnodes)

    pred_ranfomodel <- predict(ranfomodel, newdata = process_test)

    conmatrix <- table(pred_ranfomodel, changedtpye_gender_submission$Survived)

    accuracy_nonbest_ranfo <- sum(diag(conmatrix)) / sum(conmatrix)

    if (accuracy_nonbest_ranfo > setbestaccuracy) {
      setbestaccuracy <- accuracy_nonbest_ranfo
      setbestnumtree <- ntree
      setbestmaxnodes <- maxnodes
    }
  }
}

print(paste("Best number tree in random forest model =", setbestnumtree,
            "and the best max nodes =", setbestmaxnodes,
            "with the best accuracy =", setbestaccuracy))

```

```
## [1] "Best number tree in random forest model = 10 and the best max nodes = 10 with the best accuracy"
```

```

# use best n tree to do model
ranfowithoutcross <- randomForest(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,
                                  data = process_train,
                                  ntree = setbestnumtree,
                                  maxnodes = setbestmaxnodes)

pred_ranfowithoutcross <- predict(ranfowithoutcross, newdata = process_test)

```

```

conmatrix_ranfowithoutcross <- table(pred_ranfowithoutcross, changedtpye_gender_submission$Survived)

print(conmatrix_ranfowithoutcross)

##
## pred_ranfowithoutcross    0    1
##                0 260   10
##                1    6 142

TP_ranfowithoutcross <- sum(pred_ranfowithoutcross == 1 & changedtpye_gender_submission$Survived == 1)
TN_ranfowithoutcross <- sum(pred_ranfowithoutcross == 0 & changedtpye_gender_submission$Survived == 0)
FP_ranfowithoutcross <- sum(pred_ranfowithoutcross == 1 & changedtpye_gender_submission$Survived == 0)
FN_ranfowithoutcross <- sum(pred_ranfowithoutcross == 0 & changedtpye_gender_submission$Survived == 1)

cat("True Positives in Titanic RF:", TP_ranfowithoutcross, "\n")

## True Positives in Titanic RF: 142

cat("True Negatives in Titanic RF:", TN_ranfowithoutcross, "\n")

## True Negatives in Titanic RF: 260

cat("False Positives in Titanic RF:", FP_ranfowithoutcross, "\n")

## False Positives in Titanic RF: 6

cat("False Negatives in Titanic RF:", FN_ranfowithoutcross, "\n")

## False Negatives in Titanic RF: 10

table(pred_ranfowithoutcross, changedtpye_gender_submission$Survived)

##
## pred_ranfowithoutcross    0    1
##                0 260   10
##                1    6 142

acc_ranfowithoutcross <- (TP_ranfowithoutcross + TN_ranfowithoutcross) / (TP_ranfowithoutcross + TN_ranfowithoutcross)
precision_ranfowithoutcross <- (TP_ranfowithoutcross) / (TP_ranfowithoutcross + FP_ranfowithoutcross)
reca_ranfowithoutcross <- (TP_ranfowithoutcross) / (TP_ranfowithoutcross + FN_ranfowithoutcross)
f1_ranfowithoutcross <- 2 * ((precision_ranfowithoutcross * reca_ranfowithoutcross) / (precision_ranfowithoutcross + reca_ranfowithoutcross))

cat("Accuracy of Titanic RF:", acc_ranfowithoutcross, "\n")

```

```
## Accuracy of Titanic RF: 0.9617225
```

```
cat("Precision of Titanic RF:", precision_ranfowithoutcross, "\n")
```

```
## Precision of Titanic RF: 0.9594595
```

```
cat("Recall of Titanic RF:", reca_ranfowithoutcross, "\n")
```

```
## Recall of Titanic RF: 0.9342105
```

```
cat("F1-Score of Titanic RF:", f1_ranfowithoutcross, "\n")
```

```
## F1-Score of Titanic RF: 0.9466667
```

RF with k-fold cross-validation (k=10)

```
library(randomForest)

process_test$Sex <- factor(process_test$Sex, levels = levels(process_train$Sex))
process_test$Nickname <- factor(process_test$Nickname, levels = levels(process_train$Nickname))

set.seed(1)

k10 <- trainControl(method = "cv", number = 10)

#ranging from 1 to the number of predictors minus one
creatgrid <- expand.grid(mtry = 1:(ncol(process_train) - 1))

# find the best tree
samplenumtree <- c(10, 100, 250, 500, 1000)
samplemaxnodes <- c(10, 20, 30, 50, 100)

setbestnumtree <- 0
setbestmaxnodes <- 0
setbestaccuracy <- 0

for (ntree in samplenumtree) {
  for (maxnodes in samplemaxnodes) {
    ranfomodel <- train(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,
                        data = process_train,
                        method = "rf",
                        tuneGrid = creatgrid,
                        trControl = k10,
                        ntree = ntree,
                        maxnodes = maxnodes)

    pred_ranfomodel <- predict(ranfomodel, newdata = process_test)

    conmatrix <- table(pred_ranfomodel, changedtpye_gender_submission$Survived)
```

```

accuracy_nonbest_ranfo <- sum(diag(conmatrix)) / sum(conmatrix)

  if (accuracy_nonbest_ranfo > setbestaccuracy) {
    setbestaccuracy <- accuracy_nonbest_ranfo
    setbestnumtree <- ntree
    setbestmaxnodes <- maxnodes
  }
}

print(paste("Best number tree in random forest model =", setbestnumtree,
            "and the best max nodes =", setbestmaxnodes,
            "with the best accuracy =", setbestaccuracy))

```

```
## [1] "Best number tree in random forest model = 100 and the best max nodes = 10 with the best accuracy"
```

```

# use best n tree to do model
k10 <- trainControl(method = "cv", number = 10)
ranfok10 <- train(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,
                  data = process_train,
                  method = "rf",
                  tuneGrid = creatgrid,
                  trControl = k10,
                  ntree = setbestnumtree,
                  maxnodes = setbestmaxnodes
                  )

pred_ranfok10 <- predict(ranfok10, newdata = process_test)

conmatrix_ranfok10 <- table(pred_ranfok10, changedtpye_gender_submission$Survived)

print(conmatrix_ranfok10)

```

```

##
## pred_ranfok10    0    1
##                0 259    3
##                1   7 149

```

```

TP_ranfok10 <- sum(pred_ranfok10 == 1 & changedtpye_gender_submission$Survived == 1)
TN_ranfok10 <- sum(pred_ranfok10 == 0 & changedtpye_gender_submission$Survived == 0)
FP_ranfok10 <- sum(pred_ranfok10 == 1 & changedtpye_gender_submission$Survived == 0)
FN_ranfok10 <- sum(pred_ranfok10 == 0 & changedtpye_gender_submission$Survived == 1)

cat("True Positives in Titanic RF:", TP_ranfok10, "\n")

```

```
## True Positives in Titanic RF: 149
```

```

cat("True Negatives in Titanic RF:", TN_ranfok10, "\n")

## True Negatives in Titanic RF: 259

cat("False Positives in Titanic RF:", FP_ranfok10, "\n")

## False Positives in Titanic RF: 7

cat("False Negatives in Titanic RF:", FN_ranfok10, "\n")

## False Negatives in Titanic RF: 3

table(pred_ranfok10, changedtpye_gender_submission$Survived)

##
## pred_ranfok10    0    1
##               0 259    3
##               1   7 149

acc_ranfok10 <- (TP_ranfok10 + TN_ranfok10) / (TP_ranfok10 + TN_ranfok10 + FP_ranfok10 + FN_ranfok10)
precision_ranfok10 <- (TP_ranfok10) / (TP_ranfok10 + FP_ranfok10)
reca_ranfok10 <- (TP_ranfok10) / (TP_ranfok10 + FN_ranfok10)
f1_ranfok10 <- 2 * ((precision_ranfok10 * reca_ranfok10) / (precision_ranfok10 + reca_ranfok10))

cat("Accuracy of Titanic RF:", acc_ranfok10, "\n")

## Accuracy of Titanic RF: 0.9760766

cat("Precision of Titanic RF:", precision_ranfok10, "\n")

## Precision of Titanic RF: 0.9551282

cat("Recall of Titanic RF:", reca_ranfok10, "\n")

## Recall of Titanic RF: 0.9802632

cat("F1-Score of Titanic RF:", f1_ranfok10, "\n")

## F1-Score of Titanic RF: 0.9675325

```

RF with k-fold cross-validation (k=5)

```

library(randomForest)

process_test$Sex <- factor(process_test$Sex, levels = levels(process_train$Sex))
process_test$Nickname <- factor(process_test$Nickname, levels = levels(process_train$Nickname))

set.seed(1)

k5 <- trainControl(method = "cv", number = 5)

#ranging from 1 to the number of predictors minus one
creatgrid <- expand.grid(mtry = 1:(ncol(process_train) - 1))

# find the best tree
samplenumtree <- c(10, 100, 250, 500, 1000)
samplemaxnodes <- c(10, 20, 30, 50, 100)

setbestnumtree <- 0
setbestmaxnodes <- 0
setbestaccuracy <- 0

for (ntree in samplenumtree) {
  for (maxnodes in samplemaxnodes) {
    ranfomodel <- train(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,
                        data = process_train,
                        method = "rf",
                        tuneGrid = creatgrid,
                        trControl = k5,
                        ntree = ntree,
                        maxnodes = maxnodes)

    pred_ranfomodel <- predict(ranfomodel, newdata = process_test)

    conmatrix <- table(pred_ranfomodel, changedtpye_gender_submission$Survived)

    accuracy_nonbest_ranfo <- sum(diag(conmatrix)) / sum(conmatrix)

    if (accuracy_nonbest_ranfo > setbestaccuracy) {
      setbestaccuracy <- accuracy_nonbest_ranfo
      setbestnumtree <- ntree
      setbestmaxnodes <- maxnodes
    }
  }
}

# use best n tree to do model
k5 <- trainControl(method = "cv", number = 5)
ranfok5 <- train(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,
                 data = process_train,
                 method = "rf",
                 tuneGrid = creatgrid,
                 trControl = k5,
                 ntree = setbestnumtree,
                 maxnodes = setbestmaxnodes)

```

```

    )

pred_ranfok5 <- predict(ranfok5, newdata = process_test)

conmatrix_ranfok5 <- table(pred_ranfok5, changedtpye_gender_submission$Survived)

print(conmatrix_ranfok5)

##
## pred_ranfok5    0    1
##              0 256    4
##              1  10 148

TP_ranfok5 <- sum(pred_ranfok5 == 1 & changedtpye_gender_submission$Survived == 1)
TN_ranfok5 <- sum(pred_ranfok5 == 0 & changedtpye_gender_submission$Survived == 0)
FP_ranfok5 <- sum(pred_ranfok5 == 1 & changedtpye_gender_submission$Survived == 0)
FN_ranfok5 <- sum(pred_ranfok5 == 0 & changedtpye_gender_submission$Survived == 1)

cat("True Positives in Titanic RF:", TP_ranfok5, "\n")

## True Positives in Titanic RF: 148

cat("True Negatives in Titanic RF:", TN_ranfok5, "\n")

## True Negatives in Titanic RF: 256

cat("False Positives in Titanic RF:", FP_ranfok5, "\n")

## False Positives in Titanic RF: 10

cat("False Negatives in Titanic RF:", FN_ranfok5, "\n")

## False Negatives in Titanic RF: 4

table(pred_ranfok5, changedtpye_gender_submission$Survived)

##
## pred_ranfok5    0    1
##              0 256    4
##              1  10 148

acc_ranfok5 <- (TP_ranfok5 + TN_ranfok5) / (TP_ranfok5 + TN_ranfok5 + FP_ranfok5 + FN_ranfok5)

precision_ranfok5 <- (TP_ranfok5) / (TP_ranfok5 + FP_ranfok5)

reca_ranfok5 <- (TP_ranfok5) / (TP_ranfok5 + FN_ranfok5)

```

```
f1_ranfok5 <- 2 * ((precision_ranfok5 * reca_ranfok5) / (precision_ranfok5 + reca_ranfok5))

cat("Accuracy of Titanic RF:", acc_ranfok5, "\n")

## Accuracy of Titanic RF: 0.9665072

cat("Precision of Titanic RF:", precision_ranfok5, "\n")

## Precision of Titanic RF: 0.9367089

cat("Recall of Titanic RF:", reca_ranfok5, "\n")

## Recall of Titanic RF: 0.9736842

cat("F1-Score of Titanic RF:", f1_ranfok5, "\n")

## F1-Score of Titanic RF: 0.9548387
```

RF with k-fold cross-validation (k=4)

```
library(randomForest)

process_test$Sex <- factor(process_test$Sex, levels = levels(process_train$Sex))
process_test$Nickname <- factor(process_test$Nickname, levels = levels(process_train$Nickname))

set.seed(1)

k4 <- trainControl(method = "cv", number = 4)

#ranging from 1 to the number of predictors minus one
creatgrid <- expand.grid(mtry = 1:(ncol(process_train) - 1))

# find the best tree
samplenumtree <- c(10, 100, 250, 500, 1000)
samplemaxnodes <- c(10, 20, 30, 50, 100)

setbestnumtree <- 0
setbestmaxnodes <- 0
setbestaccuracy <- 0

for (ntree in samplenumtree) {
  for (maxnodes in samplemaxnodes) {
    ranfomodel <- train(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,
                        data = process_train,
                        method = "rf",
                        tuneGrid = creatgrid,
                        trControl = k4,
                        ntree = ntree,
```

```

        maxnodes = maxnodes)

pred_ranfomodel <- predict(ranfomodel, newdata = process_test)

conmatrix <- table(pred_ranfomodel, changedtpye_gender_submission$Survived)

accuracy_nonbest_ranfo <- sum(diag(conmatrix)) / sum(conmatrix)

    if (accuracy_nonbest_ranfo > setbestaccuracy) {
      setbestaccuracy <- accuracy_nonbest_ranfo
      setbestnumtree <- ntree
      setbestmaxnodes <- maxnodes
    }
  }
}

# use best n tree to do model
k4 <- trainControl(method = "cv", number = 4)
ranfok4 <- train(Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname,
                 data = process_train,
                 method = "rf",
                 tuneGrid = creatgrid,
                 trControl = k4,
                 ntree = setbestnumtree,
                 maxnodes = setbestmaxnodes
                 )

pred_ranfok4 <- predict(ranfok4, newdata = process_test)

conmatrix_ranfok4 <- table(pred_ranfok4, changedtpye_gender_submission$Survived)

print(conmatrix_ranfok4)

##
## pred_ranfok4    0    1
##               0 256    4
##               1  10 148

TP_ranfok4 <- sum(pred_ranfok4 == 1 & changedtpye_gender_submission$Survived == 1)
TN_ranfok4 <- sum(pred_ranfok4 == 0 & changedtpye_gender_submission$Survived == 0)
FP_ranfok4 <- sum(pred_ranfok4 == 1 & changedtpye_gender_submission$Survived == 0)
FN_ranfok4 <- sum(pred_ranfok4 == 0 & changedtpye_gender_submission$Survived == 1)

cat("True Positives in Titanic RF:", TP_ranfok4, "\n")

## True Positives in Titanic RF: 148

```

```

cat("True Negatives in Titanic RF:", TN_ranfok4, "\n")

## True Negatives in Titanic RF: 256

cat("False Positives in Titanic RF:", FP_ranfok4, "\n")

## False Positives in Titanic RF: 10

cat("False Negatives in Titanic RF:", FN_ranfok4, "\n")

## False Negatives in Titanic RF: 4

table(pred_ranfok4, changedtpye_gender_submission$Survived)

##
## pred_ranfok4    0    1
##              0 256    4
##              1  10 148

acc_ranfok4 <- (TP_ranfok4 + TN_ranfok4) / (TP_ranfok4 + TN_ranfok4 + FP_ranfok4 + FN_ranfok4)
precision_ranfok4 <- (TP_ranfok4) / (TP_ranfok4 + FP_ranfok4)
reca_ranfok4 <- (TP_ranfok4) / (TP_ranfok4 + FN_ranfok4)
f1_ranfok4 <- 2 * ((precision_ranfok4 * reca_ranfok4) / (precision_ranfok4 + reca_ranfok4))

cat("Accuracy of Titanic RF:", acc_ranfok4, "\n")

## Accuracy of Titanic RF: 0.9665072

cat("Precision of Titanic RF:", precision_ranfok4, "\n")

## Precision of Titanic RF: 0.9367089

cat("Recall of Titanic RF:", reca_ranfok4, "\n")

## Recall of Titanic RF: 0.9736842

cat("F1-Score of Titanic RF:", f1_ranfok4, "\n")

## F1-Score of Titanic RF: 0.9548387

```

Below is GBM-XGBoost and Neural Networks, since I need to use new features so I will using original data and start it again.

```
train <- read.csv("C:/Users/siusa/Downloads/MXN442-N11423579-Siu-Wing-Hei/train.csv")
```

```
test <- read.csv("C:/Users/siusa/Downloads/MXN442-N11423579-Siu-Wing-Hei/test.csv")
```

```
test$Name[415] <- "Oliva y Ocana, Miss. Fermina" # becuase Oliva y Ocana, Dona. Fermina do not have Nick
```

```
cat(test$Name[415])
```

```
## Oliva y Ocana, Miss. Fermina
```

```
# use mean to replace na value in column Fare
```

```
cat(test$Fare[153])
```

```
## NA
```

```
cat("\n")
```

```
test_meanofFare <- mean(test$Fare, na.rm = TRUE)
```

```
test$Fare[is.na(test$Fare)] <- test_meanofFare
```

```
cat(test$Fare[153])
```

```
## 35.62719
```

```
gender_submission <- read.csv("C:/Users/siusa/Downloads/MXN442-N11423579-Siu-Wing-Hei/gender_submission
```

Data processing

```
colnames(train)
```

```
## [1] "PassengerId" "Survived" "Pclass" "Name" "Sex"  
## [6] "Age" "SibSp" "Parch" "Ticket" "Fare"  
## [11] "Cabin" "Embarked"
```

```
str(train)
```

```
## 'data.frame': 891 obs. of 12 variables:  
## $ PassengerId: int 1 2 3 4 5 6 7 8 9 10 ...  
## $ Survived : int 0 1 1 1 0 0 0 0 1 1 ...  
## $ Pclass : int 3 1 3 1 3 3 1 3 3 2 ...  
## $ Name : chr "Braund, Mr. Owen Harris" "Cumings, Mrs. John Bradley (Florence Briggs Thayer)"  
## $ Sex : chr "male" "female" "female" "female" ...  
## $ Age : num 22 38 26 35 35 NA 54 2 27 14 ...  
## $ SibSp : int 1 1 0 1 0 0 0 3 0 1 ...  
## $ Parch : int 0 0 0 0 0 0 0 1 2 0 ...  
## $ Ticket : chr "A/5 21171" "PC 17599" "STON/O2. 3101282" "113803" ...  
## $ Fare : num 7.25 71.28 7.92 53.1 8.05 ...  
## $ Cabin : chr "" "C85" "" "C123" ...  
## $ Embarked : chr "S" "C" "S" "S" ...
```

```
str(test)
```

```
## 'data.frame': 418 obs. of 11 variables:
## $ PassengerId: int 892 893 894 895 896 897 898 899 900 901 ...
## $ Pclass : int 3 3 2 3 3 3 3 2 3 3 ...
## $ Name : chr "Kelly, Mr. James" "Wilkes, Mrs. James (Ellen Needs)" "Myles, Mr. Thomas Francis" ...
## $ Sex : chr "male" "female" "male" "male" ...
## $ Age : num 34.5 47 62 27 22 14 30 26 18 21 ...
## $ SibSp : int 0 1 0 0 1 0 0 1 0 2 ...
## $ Parch : int 0 0 0 0 1 0 0 1 0 0 ...
## $ Ticket : chr "330911" "363272" "240276" "315154" ...
## $ Fare : num 7.83 7 9.69 8.66 12.29 ...
## $ Cabin : chr "" "" "" "" ...
## $ Embarked : chr "Q" "S" "Q" "S" ...
```

```
str(gender_submission)
```

```
## 'data.frame': 418 obs. of 2 variables:
## $ PassengerId: int 892 893 894 895 896 897 898 899 900 901 ...
## $ Survived : int 0 1 0 0 1 0 1 0 1 0 ...
```

```
changedtpye_train<-data.frame(PassengerId = train$PassengerId,
                               Survived = as.factor(train$Survived),
                               Pclass = as.factor(train$Pclass),
                               Name = train$Name,
                               Sex = as.factor(train$Sex),
                               Age = train$Age,
                               SibSp = train$SibSp,
                               Parch = train$Parch,
                               Ticket = train$Ticket,
                               Fare = train$Fare,
                               Cabin = train$Cabin,
                               Embarked = train$Embarked
                               )
```

```
str(changedtpye_train)
```

```
## 'data.frame': 891 obs. of 12 variables:
## $ PassengerId: int 1 2 3 4 5 6 7 8 9 10 ...
## $ Survived : Factor w/ 2 levels "0","1": 1 2 2 2 1 1 1 1 2 2 ...
## $ Pclass : Factor w/ 3 levels "1","2","3": 3 1 3 1 3 3 1 3 3 2 ...
## $ Name : chr "Braund, Mr. Owen Harris" "Cumings, Mrs. John Bradley (Florence Briggs Thayer)" ...
## $ Sex : Factor w/ 2 levels "female","male": 2 1 1 1 2 2 2 2 1 1 ...
## $ Age : num 22 38 26 35 35 NA 54 2 27 14 ...
## $ SibSp : int 1 1 0 1 0 0 0 3 0 1 ...
## $ Parch : int 0 0 0 0 0 0 0 1 2 0 ...
## $ Ticket : chr "A/5 21171" "PC 17599" "STON/O2. 3101282" "113803" ...
## $ Fare : num 7.25 71.28 7.92 53.1 8.05 ...
## $ Cabin : chr "" "C85" "" "C123" ...
## $ Embarked : chr "S" "C" "S" "S" ...
```

```

changedtpye_test<-data.frame(PassengerId = test$PassengerId,
                             Pclass = as.factor(test$Pclass),
                             Name = test$Name,
                             Sex = as.factor(test$Sex),
                             Age = test$Age,
                             SibSp = test$SibSp,
                             Parch = test$Parch,
                             Ticket = test$Ticket,
                             Fare = test$Fare,
                             Cabin = test$Cabin,
                             Embarked = test$Embarked
                             )

str(changedtpye_test)

```

```

## 'data.frame':    418 obs. of  11 variables:
## $ PassengerId: int  892 893 894 895 896 897 898 899 900 901 ...
## $ Pclass     : Factor w/ 3 levels "1","2","3": 3 3 2 3 3 3 3 2 3 3 ...
## $ Name       : chr  "Kelly, Mr. James" "Wilkes, Mrs. James (Ellen Needs)" "Myles, Mr. Thomas Francis" ...
## $ Sex        : Factor w/ 2 levels "female","male": 2 1 2 2 1 2 1 2 1 2 ...
## $ Age        : num  34.5 47 62 27 22 14 30 26 18 21 ...
## $ SibSp      : int   0 1 0 0 1 0 0 1 0 2 ...
## $ Parch      : int   0 0 0 0 1 0 0 1 0 0 ...
## $ Ticket     : chr   "330911" "363272" "240276" "315154" ...
## $ Fare       : num   7.83 7 9.69 8.66 12.29 ...
## $ Cabin      : chr    "" "" "" "" ...
## $ Embarked   : chr   "Q" "S" "Q" "S" ...

```

```

changedtpye_gender_submission<-data.frame(PassengerId = gender_submission$PassengerId,
                                           Survived = as.factor(gender_submission$Survived)
                                           )

str(changedtpye_gender_submission)

```

```

## 'data.frame':    418 obs. of  2 variables:
## $ PassengerId: int  892 893 894 895 896 897 898 899 900 901 ...
## $ Survived   : Factor w/ 2 levels "0","1": 1 2 1 1 2 1 2 1 2 1 ...

```

chagne begging here

```

# delet row passengerID
process_train <- subset(changedtpye_train, select = -PassengerId)

# change age to mean
train_meanofAge <- mean(process_train$Age, na.rm = TRUE)
process_train$Age[is.na(process_train$Age)] <- train_meanofAge

# make change in name
process_train$Nickname <- sub(".*\\s*(\\w+)\\.\\.\\s*", "\\1", process_train$Name)
process_train$Nickname <- as.factor(process_train$Nickname)

```

```
process_train <- subset(process_train, select = -Name)#must do this after already change Nickname

# keep below columns, different to Artilce

#process_train <- subset(process_train, select = -Ticket)
#process_train <- subset(process_train, select = -Fare)
#process_train <- subset(process_train, select = -Cabin)
#process_train <- subset(process_train, select = -Embarked)

head(process_train)
```

##	Survived	Pclass	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin
## 1	0	3	male	22.00000	1	0	A/5 21171	7.2500	
## 2	1	1	female	38.00000	1	0	PC 17599	71.2833	C85
## 3	1	3	female	26.00000	0	0	STON/O2. 3101282	7.9250	
## 4	1	1	female	35.00000	1	0	113803	53.1000	C123
## 5	0	3	male	35.00000	0	0	373450	8.0500	
## 6	0	3	male	29.69912	0	0	330877	8.4583	

##	Embarked	Nickname
## 1	S	Mr
## 2	C	Mrs
## 3	S	Miss
## 4	S	Mrs
## 5	S	Mr
## 6	Q	Mr

```
# delet row passengerID
process_test <- subset(changedtpye_test, select = -PassengerId)

# change age to mean
test_meanofAge <- mean(process_test$Age, na.rm = TRUE)
process_test$Age[is.na(process_test$Age)] <- test_meanofAge

# make change in name
process_test$Nickname <- sub(".*,(\\s*(\\w+))\\.\\s*", "\\1", process_test$Name)
process_test$Nickname <- as.factor(process_test$Nickname)
#process_test$Nickname[415] <- "Miss" # becuase Oliva y Ocana, Dona. Fermina do not have Nickname

process_test <- subset(process_test, select = -Name)#must do this after already change Nickname

# keep below columns, different to Artilce

#process_test <- subset(process_test, select = -Ticket)
#process_test <- subset(process_test, select = -Fare)
#process_test <- subset(process_test, select = -Cabin)
#process_test <- subset(process_test, select = -Embarked)
head(process_test)
```

##	Pclass	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	Nickname
## 1	3	male	34.5	0	0	330911	7.8292		Q	Mr

```
## 2      3 female 47.0      1      0 363272 7.0000      S      Mrs
## 3      2  male 62.0      0      0 240276 9.6875      Q      Mr
## 4      3  male 27.0      0      0 315154 8.6625      S      Mr
## 5      3 female 22.0      1      1 3101298 12.2875     S      Mrs
## 6      3  male 14.0      0      0   7538 9.2250      S      Mr
```

```
library(caret)
library(e1071)
library(randomForest)
library(class)
library(dplyr)
library(rpart)
```

Gradient Boosting Machine-Extreme Gradient Boosting

GBM-XGBoost with out cross-validation

```
library(xgboost)
library(caret)
library(Matrix)
```

```
table(process_train$Survived)
```

```
##
##      0      1
## 549 342
```

```
# Calculate the ratio of the classes
prop.table(table(process_train$Survived))
```

```
##
##           0           1
## 0.6161616 0.3838384
```

```
library(xgboost)
library(caret)
library(Matrix)
```

```
process_test$Sex <- factor(process_test$Sex, levels = levels(process_train$Sex))
process_test$Nickname <- factor(process_test$Nickname, levels = levels(process_train$Nickname))
```

```
set.seed(1)
```

```
allpreicter <- model.matrix(Survived ~ Pclass + Nickname + Sex + Age + SibSp + Parch + Ticket + Fare +
```

```
process_train$Survived <- factor(process_train$Survived, levels = c("0", "1"))
dtypenum_survived0or1 <- as.numeric(as.character(process_train$Survived))
```

```
TitanicXGBoosttrain <- xgb.DMatrix(data = as.matrix(allpreicter), label = dtypenum_survived0or1)
```

```

set.seed(1)
parofTitanicXG <- list(
  booster = "gbtree",
  objective = "binary:logistic",
  #eval_metric = c("error", "auc", "logloss"),
  eval_metric = c("error"),
  eval_metric = c("auc"),
  eval_metric = c("logloss"),
  # classification error rate and area under curve and negative log-likelihood
  eta = 0.1, # small value training time decrease
  max_depth = 3, # less value-less complex-less overfitting
  subsample = 0.8, # my training set subsample rate
  colsample_bytree = 0.8,
  scale_pos_weight = 1.6047,
  # Titanic train data set is not survived, 549 (survived)/ 342(not survived) = 1.6047
  gamma = 1,
  lambda = 1,
  alpha = 0
)

MyTitanicrounds = 100
stophere = 5

XGwithoutcross <- xgb.train(
  params = parofTitanicXG,
  data = TitanicXGBoosttrain,
  nrounds = MyTitanicrounds,
  early_stopping_rounds = stophere,
  watchlist = list(train = TitanicXGBoosttrain),
  verbose = 0
)

```

Can not do Confusion Matrix if use all preictors. Since have column value is very different, of example Ticket, Cabin each value of example PC 17608(Ticket), B57 B59 B63 B66(Cabin) is different to each passagers, so cannot do Confusion Matrix in all preictors.

Embarked also cannot use in Confusion Matrix.

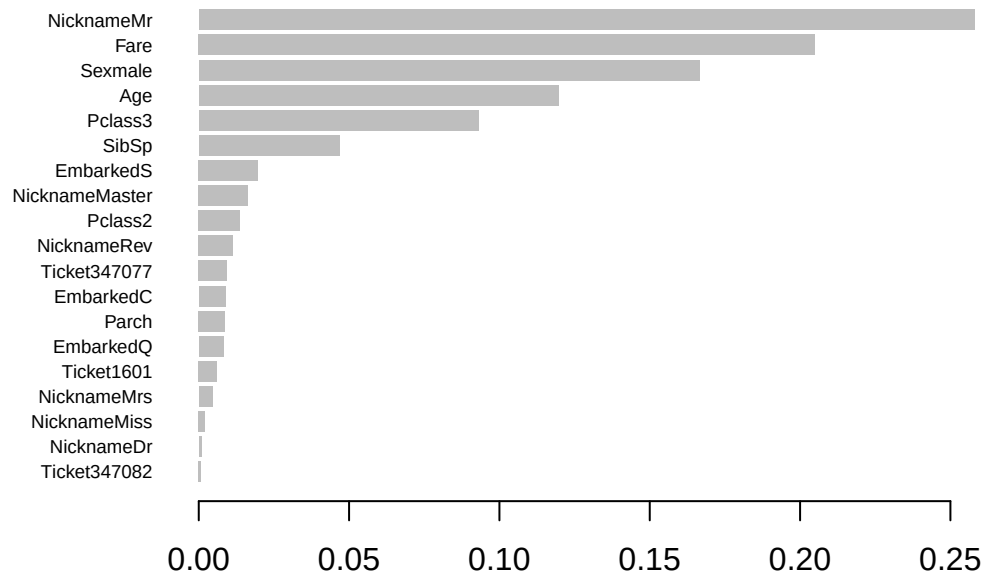
So need to use variable selection, it also mean, On the surface even use all preictors performance is better than after select preictors, but auctually select preictors is better.

```
# can call Importance-based Selection or
# or Importance-based Feature Selection or Filter-based Feature Selection
selectXGwithoutcross <- xgb.importance(feature_names = colnames(allpreicter), model = XGwithoutcross)

print(selectXGwithoutcross)
```

##	Feature	Gain	Cover	Frequency
##	<char>	<num>	<num>	<num>
## 1:	NicknameMr	0.2579617405	0.0617910006	0.032136106
## 2:	Fare	0.2050165868	0.2923827038	0.317580340
## 3:	Sexmale	0.1664666780	0.0753474383	0.056710775
## 4:	Age	0.1196055475	0.2347019815	0.255198488
## 5:	Pclass3	0.0931158534	0.0713015828	0.068052930
## 6:	SibSp	0.0468941417	0.0564534207	0.069943289
## 7:	EmbarkedS	0.0196977082	0.0232732464	0.030245747
## 8:	NicknameMaster	0.0164506309	0.0360718085	0.024574669
## 9:	Pclass2	0.0137269420	0.0248664312	0.024574669
## 10:	NicknameRev	0.0113629289	0.0078062957	0.009451796
## 11:	Ticket347077	0.0093523719	0.0366389393	0.022684310
## 12:	EmbarkedC	0.0089184800	0.0191051564	0.018903592
## 13:	Parch	0.0087289181	0.0217682718	0.024574669
## 14:	EmbarkedQ	0.0082632759	0.0055330790	0.011342155
## 15:	Ticket1601	0.0061057447	0.0100657896	0.009451796
## 16:	NicknameMrs	0.0046085087	0.0177979548	0.013232514
## 17:	NicknameMiss	0.0020416615	0.0023370858	0.007561437
## 18:	NicknameDr	0.0009400701	0.0009549138	0.001890359
## 19:	Ticket347082	0.0007422113	0.0018029001	0.001890359

```
xgb.plot.importance(selectXGwithoutcross)
```



GBM-XGBoost with out cross-validation

```
process_test$Sex <- factor(process_test$Sex, levels = levels(process_train$Sex))
process_test$Nickname <- factor(process_test$Nickname, levels = levels(process_train$Nickname))

set.seed(1)
allpreicter <- model.matrix(Survived ~ Pclass + Sex + Age + SibSp + Fare + Nickname, data = process_train)

process_train$Survived <- factor(process_train$Survived, levels = c("0", "1"))
dtypenum_survived0or1 <- as.numeric(as.character(process_train$Survived))

TitanicXGBoosttrain <- xgb.DMatrix(data = as.matrix(allpreicter), label = dtypenum_survived0or1)

set.seed(1)
parofTitanicXG <- list(
  booster = "gbtree",
  objective = "binary:logistic",
  #eval_metric = c("error", "auc", "logloss"),
  eval_metric = c("error"),
  eval_metric = c("auc"),
  eval_metric = c("logloss"),
```

```

# classification error rate and area under curve and negative log-likelihood
eta = 0.1, # small value training time decrease
max_depth = 3, # less value-less complex-less overfitting
subsample = 0.8, # my training set subsample rate
colsample_bytree = 0.8,
scale_pos_weight = 1.6047,
# Titanic train data set is not survived, 549 (survived)/ 342(not survived) = 1.6047
gamma = 1,
lambda = 1,
alpha = 0
)

MyTitanicrounds = 100
stophere = 5

XGwithoutcross <- xgb.train(
  params = parofTitanicXG,
  data = TitanicXGBoosttrain,
  nrounds = MyTitanicrounds,
  early_stopping_rounds = stophere,
  watchlist = list(train = TitanicXGBoosttrain),
  verbose = 0
)

allpreict_test <- model.matrix(~ Pclass + Sex + Age + SibSp + Fare + Nickname, data = process_test)
TitanicXGBoosttest <- xgb.DMatrix(data = as.matrix(allpreict_test))

pre_XGwithoutcross <- predict(XGwithoutcross, newdata = TitanicXGBoosttest)

pre_value_XGwithoutcross <- ifelse(pre_XGwithoutcross > 0.5, 1, 0)

conmatrix_XGwithoutcross <- table(pre_value_XGwithoutcross, changedtpye_gender_submission$Survived)
print(conmatrix_XGwithoutcross)

##
## pre_value_XGwithoutcross    0    1
##                0 237   14
##                1   29  138

TP_XGwithoutcross <- sum(pre_value_XGwithoutcross == 1 & changedtpye_gender_submission$Survived == 1)
TN_XGwithoutcross <- sum(pre_value_XGwithoutcross == 0 & changedtpye_gender_submission$Survived == 0)
FP_XGwithoutcross <- sum(pre_value_XGwithoutcross == 1 & changedtpye_gender_submission$Survived == 0)
FN_XGwithoutcross <- sum(pre_value_XGwithoutcross == 0 & changedtpye_gender_submission$Survived == 1)

```

```
cat("True Positives in Titanic XGBoost (TP):", TP_XGwithoutcross, "\n")
```

```
## True Positives in Titanic XGBoost (TP): 138
```

```
cat("True Negatives in Titanic XGBoost (TN):", TN_XGwithoutcross, "\n")
```

```
## True Negatives in Titanic XGBoost (TN): 237
```

```
cat("False Positives in Titanic XGBoost (FP):", FP_XGwithoutcross, "\n")
```

```
## False Positives in Titanic XGBoost (FP): 29
```

```
cat("False Negatives in Titanic XGBoost (FN):", FN_XGwithoutcross, "\n")
```

```
## False Negatives in Titanic XGBoost (FN): 14
```

```
acc_XGwithoutcross <- (TP_XGwithoutcross + TN_XGwithoutcross) / (TP_XGwithoutcross + TN_XGwithoutcross + FP_XGwithoutcross + FN_XGwithoutcross)
```

```
precision_XGwithoutcross <- (TP_XGwithoutcross) / (TP_XGwithoutcross + FP_XGwithoutcross)
```

```
reca_XGwithoutcross <- (TP_XGwithoutcross) / (TP_XGwithoutcross + FN_XGwithoutcross)
```

```
f1_XGwithoutcross <- 2 * ((precision_XGwithoutcross * reca_XGwithoutcross) / (precision_XGwithoutcross + reca_XGwithoutcross))
```

```
cat("Accuracy of Titanic XGBoost:", acc_XGwithoutcross, "\n")
```

```
## Accuracy of Titanic XGBoost: 0.8971292
```

```
cat("Precision of Titanic XGBoost:", precision_XGwithoutcross, "\n")
```

```
## Precision of Titanic XGBoost: 0.8263473
```

```
cat("Recall of Titanic XGBoost:", reca_XGwithoutcross, "\n")
```

```
## Recall of Titanic XGBoost: 0.9078947
```

```
cat("F1-Score of Titanic XGBoost:", f1_XGwithoutcross, "\n")
```

```
## F1-Score of Titanic XGBoost: 0.8652038
```

GBM-XGBoost with k-fold cross-validation (k=10)

```

origprocess_train <- process_train
origprocess_test <- process_test
process_test$Sex <- factor(process_test$Sex, levels = levels(process_train$Sex))
process_test$Nickname <- factor(process_test$Nickname, levels = levels(process_train$Nickname))

set.seed(1)
allpreicter <- model.matrix(Survived ~ Pclass + Sex + Age + SibSp + Fare + Nickname, data = process_train)
process_train$Survived <- factor(process_train$Survived, levels = c("0", "1"))
dtypenum_survived0or1 <- as.numeric(as.character(process_train$Survived))
TitanicXGBoosttrain <- xgb.DMatrix(data = as.matrix(allpreicter), label = dtypenum_survived0or1)

set.seed(1)
parofTitanicXG <- list(
  booster = "gbtree",
  objective = "binary:logistic",
  #eval_metric = c("error", "auc", "logloss"),
  eval_metric = c("error"),
  eval_metric = c("auc"),
  eval_metric = c("logloss"),
  # classification error rate and area under curve and negative log-likelihood
  eta = 0.1, # small value training time decrease
  max_depth = 3, # less value-less complex-less overfitting
  subsample = 0.8, # my training set subsample rate
  colsample_bytree = 0.8,
  scale_pos_weight = 1.6047,
  # Titanic train data set is not survived, 549 (survived)/ 342(not survived) = 1.6047
  gamma = 1,
  lambda = 1,
  alpha = 0
)

MyTitanicrounds = 100
stophere = 5
k10 <- 10

set.seed(1)
findbest <- xgb.cv(
  params = parofTitanicXG,
  data = TitanicXGBoosttrain,
  nrounds = MyTitanicrounds,
  nfold = k10,
  early_stopping_rounds = stophere,
  verbose = 0
)

# Get the best number of boosting rounds from k=10cross-validation
Titbesnro <- findbest$best_iteration

# Train it again since we have the best n rounds already
XGwithk10Cross <- xgb.train(
  params = parofTitanicXG,
  data = TitanicXGBoosttrain,

```

```

nrounds = Titbesnro,
watchlist = list(train = TitanicXGBoosttrain),
verbose = 0
)

allpreict_test <- model.matrix(~ Pclass + Sex + Age + SibSp + Fare + Nickname, data = process_test)
TitanicXGBoosttest <- xgb.DMatrix(data = as.matrix(allpreict_test))

pre_XGwithk10Cross <- predict(XGwithk10Cross, newdata = TitanicXGBoosttest)

pre_value_XGwithk10Cross <- ifelse(pre_XGwithk10Cross > 0.5, 1, 0)

conmatrix_XGwithk10Cross <- table(pre_value_XGwithk10Cross, changedtpye_gender_submission$Survived)
print(conmatrix_XGwithk10Cross)

##
## pre_value_XGwithk10Cross    0    1
##                          0 235    6
##                          1  31 146

process_train <- origprocess_train
process_test <- origprocess_test

TP_XGwithk10Cross <- sum(pre_value_XGwithk10Cross == 1 & changedtpye_gender_submission$Survived == 1)
TN_XGwithk10Cross <- sum(pre_value_XGwithk10Cross == 0 & changedtpye_gender_submission$Survived == 0)
FP_XGwithk10Cross <- sum(pre_value_XGwithk10Cross == 1 & changedtpye_gender_submission$Survived == 0)
FN_XGwithk10Cross <- sum(pre_value_XGwithk10Cross == 0 & changedtpye_gender_submission$Survived == 1)

cat("True Positives in Titanic XGBoost (TP):", TP_XGwithk10Cross, "\n")

## True Positives in Titanic XGBoost (TP): 146

cat("True Negatives in Titanic XGBoost (TN):", TN_XGwithk10Cross, "\n")

## True Negatives in Titanic XGBoost (TN): 235

cat("False Positives in Titanic XGBoost (FP):", FP_XGwithk10Cross, "\n")

## False Positives in Titanic XGBoost (FP): 31

cat("False Negatives in Titanic XGBoost (FN):", FN_XGwithk10Cross, "\n")

## False Negatives in Titanic XGBoost (FN): 6

```

```
acc_XGwithk10Cross <- (TP_XGwithk10Cross + TN_XGwithk10Cross) / (TP_XGwithk10Cross + TN_XGwithk10Cross + FP_XGwithk10Cross + FN_XGwithk10Cross)
precision_XGwithk10Cross <- TP_XGwithk10Cross / (TP_XGwithk10Cross + FP_XGwithk10Cross)
recall_XGwithk10Cross <- TP_XGwithk10Cross / (TP_XGwithk10Cross + FN_XGwithk10Cross)
f1_XGwithk10Cross <- 2 * ((precision_XGwithk10Cross * recall_XGwithk10Cross) / (precision_XGwithk10Cross + recall_XGwithk10Cross))

cat("Accuracy of Titanic XGBoost:", acc_XGwithk10Cross, "\n")
```

```
## Accuracy of Titanic XGBoost: 0.9114833
```

```
cat("Precision of Titanic XGBoost:", precision_XGwithk10Cross, "\n")
```

```
## Precision of Titanic XGBoost: 0.8248588
```

```
cat("Recall of Titanic XGBoost:", recall_XGwithk10Cross, "\n")
```

```
## Recall of Titanic XGBoost: 0.9605263
```

```
cat("F1-Score of Titanic XGBoost:", f1_XGwithk10Cross, "\n")
```

```
## F1-Score of Titanic XGBoost: 0.887538
```

Neural Networks

Neural Networks with out cross-validation

I can not do variable selection in Neural Networks since some high-cardinality categorical variables like cabin, make code can not run.

So I decide to use Random FOrrest to do variable selection. Random Forest do well with binary issue because it builds decision trees that split based on features values

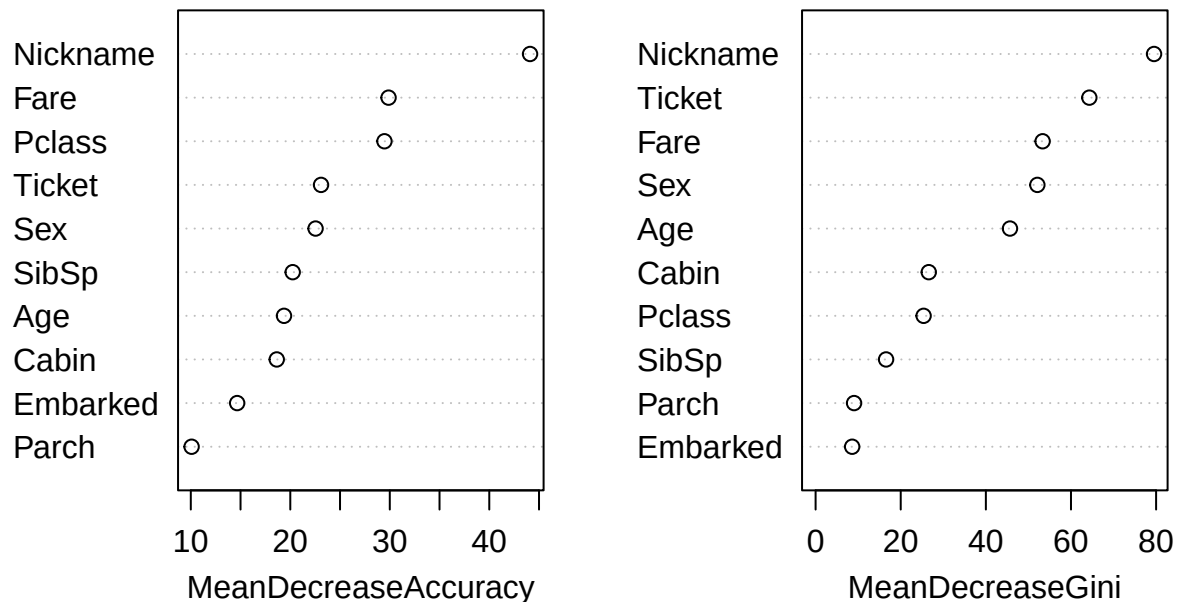
```
library(randomForest)
set.seed(1)
featuresselectionuserf <- randomForest(Survived ~ Pclass + Nickname + Sex + Age + SibSp + Parch + Ticket, data=train, importance=TRUE)

# View variable importance
importance(featuresselectionuserf)
```

	0	1	MeanDecreaseAccuracy	MeanDecreaseGini
## Pclass	19.938770	21.623345	29.46355	25.374447
## Nickname	39.830817	21.750997	44.09922	79.517709
## Sex	22.790088	7.483800	22.54230	52.102628
## Age	13.220210	14.620248	19.36865	45.667594
## SibSp	20.367184	2.374422	20.24087	16.556233
## Parch	7.971105	5.540939	10.08153	9.037348
## Ticket	18.206845	11.445976	23.09219	64.311089
## Fare	13.316875	25.391037	29.87746	53.335764
## Cabin	16.119194	8.528915	18.63930	26.586801
## Embarked	10.437984	9.785090	14.66273	8.590934

```
# Plot variable importance
varImpPlot(featureselectionuserf)
```

featureselectionuserf



We need to ignore Ticket, Cabin and Embarked.

In the Gini diagram, the features Nickname, Fare, Sex, and Age show strong contributions to classification performance. The Accuracy diagram also indicates that Nickname, Fare, and Pclass perform well. Considering two diagram, we should prioritize the following features: Nickname, Fare, Pclass, Sex, SibSp, and Age in the model.

I Neural Networks I will search which group of features is the best of accuracy (Nickname, Fare, Pclass, Sex, SibSp, and Age), and find which situation provide the best result (Hidden layer from 1 to 10, decay 0.01 or 0.01 or 0.1, maxit 100 or 200)

```
# single layer feedforward neural network
library(nnet)

newfeatures <- c("Nickname", "Fare", "Sex", "Age", "Pclass", "SibSP")
Hiddenlayer1to10 <- 1:10 # one hidden layer with 1 to 10 neurons
overfi <- c(0.01, 0.01, 0.1)
converge <- c(100, 200)

nnbesacc <- 0
```

```

lastbes <- list()

for (i in 1:length(newfeatures)) {
  current_newfeatures <- newfeatures[1:i]
  dostru <- paste("Survived ~", paste(current_newfeatures, collapse = " + "))
  newfeaturesformu <- as.formula(dostru)

  for (size in Hiddenlayer1to10) {
    for (decay in overfi) {
      for (maxit in converge) {
        tryCatch({
          set.seed(1)
          nnwithoutcross <- nnet(newfeaturesformu, data = process_train, size = size,
                                trace = FALSE, maxit = maxit, decay = decay)

          pre_nnwithoutcross <- predict(nnwithoutcross, newdata = process_test, type = "class")
          TP_nnwithoutcross <- sum(pre_nnwithoutcross == 1 & changedtpye_gender_submission$Survived == 1)
          TN_nnwithoutcross <- sum(pre_nnwithoutcross == 0 & changedtpye_gender_submission$Survived == 0)
          FP_nnwithoutcross <- sum(pre_nnwithoutcross == 1 & changedtpye_gender_submission$Survived == 0)
          FN_nnwithoutcross <- sum(pre_nnwithoutcross == 0 & changedtpye_gender_submission$Survived == 1)
          acc_nnwithoutcross <- (TP_nnwithoutcross + TN_nnwithoutcross) /
            (TP_nnwithoutcross + TN_nnwithoutcross + FP_nnwithoutcross + FN_nnwithoutcross)

          if (acc_nnwithoutcross > nnbesacc) {
            nnbesacc <- acc_nnwithoutcross
            lastbes <- list(newfeatureset = current_newfeatures, size = size, decay = decay,
                          accuracy = acc_nnwithoutcross, maxit = maxit
            )
          }
        }, error = function(e) {
        })
      }
    }
  }
}

cat("The best feature set in Titanic NN model:", paste(lastbes$newfeatureset, collapse = " + "), "\n")

## The best feature set in Titanic NN model: Nickname + Fare

cat("Best hidden layer size of NN Titanic:", lastbes$size, "\n")

## Best hidden layer size of NN Titanic: 2

cat("Best decay value of NN Titanic:", lastbes$decay, "\n")

## Best decay value of NN Titanic: 0.01

cat("Best maxit value of NN Titanic:", lastbes$maxit, "\n")

## Best maxit value of NN Titanic: 100

```

```
cat("Highest Accuracy of NN Titanic:", lastbes$accuracy, "\n")
```

```
## Highest Accuracy of NN Titanic: 0.9665072
```

```
#in above code we find Nickname + Fare , 2size, 0.01decay 100maxit is the best  
set.seed(1)
```

```
nnwithoutcross <- nnet(Survived ~ Nickname + Fare,  
  data = process_train,  
  size = 2, # Adjust size as needed  
  decay = 0.01,  
  maxit = 100)
```

```
## # weights: 39  
## initial value 634.826613  
## iter 10 value 532.241474  
## iter 20 value 431.857554  
## iter 30 value 424.748256  
## iter 40 value 423.613699  
## iter 50 value 422.599059  
## iter 60 value 422.292789  
## iter 70 value 416.687062  
## iter 80 value 414.426109  
## iter 90 value 413.448793  
## iter 100 value 412.778595  
## final value 412.778595  
## stopped after 100 iterations
```

```
pre_nnwithoutcross <- predict(nnwithoutcross, newdata = process_test, type = "class")
```

```
conmatrix_nnwithoutcross <- table(pre_nnwithoutcross, changedtpye_gender_submission$Survived)
```

```
print(conmatrix_nnwithoutcross)
```

```
##  
## pre_nnwithoutcross    0    1  
##           0 252    0  
##           1  14 152
```

```
TP_nnwithoutcross <- sum(pre_nnwithoutcross == 1 & changedtpye_gender_submission$Survived == 1)  
TN_nnwithoutcross <- sum(pre_nnwithoutcross == 0 & changedtpye_gender_submission$Survived == 0)  
FP_nnwithoutcross <- sum(pre_nnwithoutcross == 1 & changedtpye_gender_submission$Survived == 0)  
FN_nnwithoutcross <- sum(pre_nnwithoutcross == 0 & changedtpye_gender_submission$Survived == 1)
```

```
cat("True Positives in Titanic NN (TP):", TP_nnwithoutcross, "\n")
```

```
## True Positives in Titanic NN (TP): 152
```

```
cat("True Negatives in Titanic NN (TN):", TN_nnwithoutcross, "\n")
```

```
## True Negatives in Titanic NN (TN): 252
```

```
cat("False Positives in Titanic NN (FP):", FP_nnwithoutcross, "\n")
```

```
## False Positives in Titanic NN (FP): 14
```

```
cat("False Negatives in Titanic NN (FN):", FN_nnwithoutcross, "\n")
```

```
## False Negatives in Titanic NN (FN): 0
```

```
acc_nnwithoutcross <- (TP_nnwithoutcross + TN_nnwithoutcross) / (TP_nnwithoutcross + TN_nnwithoutcross + FP_nnwithoutcross + FN_nnwithoutcross)
precision_nnwithoutcross <- TP_nnwithoutcross / (TP_nnwithoutcross + FP_nnwithoutcross)
recall_nnwithoutcross <- TP_nnwithoutcross / (TP_nnwithoutcross + FN_nnwithoutcross)
f1_nnwithoutcross <- 2 * ((precision_nnwithoutcross * recall_nnwithoutcross) / (precision_nnwithoutcross + recall_nnwithoutcross))

cat("Accuracy of Titanic NN:", acc_nnwithoutcross, "\n")
```

```
## Accuracy of Titanic NN: 0.9665072
```

```
cat("Precision of Titanic NN:", precision_nnwithoutcross, "\n")
```

```
## Precision of Titanic NN: 0.9156627
```

```
cat("Recall of Titanic NN:", recall_nnwithoutcross, "\n")
```

```
## Recall of Titanic NN: 1
```

```
cat("F1-Score of Titanic NN:", f1_nnwithoutcross, "\n")
```

```
## F1-Score of Titanic NN: 0.9559748
```

Neural Networks with k-fold cross-validation (k=10)

```
# single layer feedforward neural network
library(nnet)
library(caret)

k10 <- trainControl(method = "cv", number = 10)

newfeatures <- c("Nickname", "Fare", "Sex", "Age", "Pclass", "SibSP")
Hiddenlayer1to10 <- 1:10 # one hidden layer with 1 to 10 neurons
overfi <- c(0.01, 0.01, 0.1)
converge <- c(100, 200)

nnbesacc <- 0
lastbes <- list()

for (i in 1:length(newfeatures)) {
```

```

current_newfeatures <- newfeatures[1:i]
dostru <- paste("Survived ~", paste(current_newfeatures, collapse = " + "))
newfeaturesformu <- as.formula(dostru)

for (size in Hiddenlayer1to10) {
  for (decay in overfi) {
    for (maxit in converge) {
      tryCatch({
        set.seed(1)

        nnwithk10 <- train(newfeaturesformu,
                           data = process_train,
                           method = "nnet",
                           trControl = k10,
                           tuneGrid = expand.grid(.size = size, .decay = decay),
                           trace = FALSE,
                           maxit = maxit)

        pre_nnwithk10 <- predict(nnwithk10, newdata = process_test)

        TP_nnwithk10 <- sum(pre_nnwithk10 == 1 & changedtpye_gender_submission$Survived == 1)
        TN_nnwithk10 <- sum(pre_nnwithk10 == 0 & changedtpye_gender_submission$Survived == 0)
        FP_nnwithk10 <- sum(pre_nnwithk10 == 1 & changedtpye_gender_submission$Survived == 0)
        FN_nnwithk10 <- sum(pre_nnwithk10 == 0 & changedtpye_gender_submission$Survived == 1)

        acc_nnwithk10 <- (TP_nnwithk10 + TN_nnwithk10) /
          (TP_nnwithk10 + TN_nnwithk10 + FP_nnwithk10 + FN_nnwithk10)

        if (acc_nnwithk10 > nnbesacc) {
          nnbesacc <- acc_nnwithk10
          lastbes <- list(newfeatureset = current_newfeatures, size = size, decay = decay,
                        accuracy = acc_nnwithk10, maxit = maxit
          )
        }
      }, error = function(e) {
      })
    }
  }
}

cat("The best feature set in Titanic NN model:", paste(lastbes$newfeatureset, collapse = " + "), "\n")

## The best feature set in Titanic NN model: Nickname + Fare

cat("Best hidden layer size of NN Titanic:", lastbes$size, "\n")

## Best hidden layer size of NN Titanic: 3

cat("Best decay value of NN Titanic:", lastbes$decay, "\n")

```

```
## Best decay value of NN Titanic: 0.01
```

```
cat("Best maxit value of NN Titanic:", lastbes$maxit, "\n")
```

```
## Best maxit value of NN Titanic: 200
```

```
cat("Highest Accuracy of NN Titanic:", lastbes$accuracy, "\n")
```

```
## Highest Accuracy of NN Titanic: 0.9665072
```

```
# in k = 10 we find Nickname + Fare , 3size, 0.01decay 200maxit is the best  
set.seed(1)
```

```
nnwithk10 <- train(Survived ~ Nickname + Fare, data = process_train, method = "nnet",  
  trControl = k10, tuneGrid = expand.grid(.size = 3, .decay = 0.01),  
  trace = FALSE, maxit = 200)
```

```
pre_nnwithk10 <- predict(nnwithk10, newdata = process_test, type = "raw")
```

```
conmatrix_nnwithk10 <- table(pre_nnwithk10, changedtpye_gender_submission$Survived)
```

```
print(conmatrix_nnwithk10)
```

```
##
```

```
## pre_nnwithk10    0    1
```

```
##                0 252    0
```

```
##                1  14 152
```

```
TP_nnwithk10 <- sum(pre_nnwithk10 == 1 & changedtpye_gender_submission$Survived == 1)
```

```
TN_nnwithk10 <- sum(pre_nnwithk10 == 0 & changedtpye_gender_submission$Survived == 0)
```

```
FP_nnwithk10 <- sum(pre_nnwithk10 == 1 & changedtpye_gender_submission$Survived == 0)
```

```
FN_nnwithk10 <- sum(pre_nnwithk10 == 0 & changedtpye_gender_submission$Survived == 1)
```

```
cat("True Positives in Titanic NN:", TP_nnwithk10, "\n")
```

```
## True Positives in Titanic NN: 152
```

```
cat("True Negatives in Titanic NN:", TN_nnwithk10, "\n")
```

```
## True Negatives in Titanic NN: 252
```

```
cat("False Positives in Titanic NN:", FP_nnwithk10, "\n")
```

```
## False Positives in Titanic NN: 14
```

```
cat("False Negatives in Titanic NN:", FN_nnwithk10, "\n")
```

```
## False Negatives in Titanic NN: 0
```

```
table(pre_nnwithk10, changedtpye_gender_submission$Survived)
```

```
##  
## pre_nnwithk10    0    1  
##              0 252    0  
##              1  14 152
```

```
acc_nnwithk10 <- (TP_nnwithk10 + TN_nnwithk10) / (TP_nnwithk10 + TN_nnwithk10 + FP_nnwithk10 + FN_nnwithk10)
```

```
precision_nnwithk10 <- (TP_nnwithk10) / (TP_nnwithk10 + FP_nnwithk10)
```

```
reca_nnwithk10 <- (TP_nnwithk10) / (TP_nnwithk10 + FN_nnwithk10)
```

```
f1_nnwithk10 <- 2 * ((precision_nnwithk10 * reca_nnwithk10) / (precision_nnwithk10 + reca_nnwithk10))
```

```
cat("Accuracy of Titanic NN:", acc_nnwithk10, "\n")
```

```
## Accuracy of Titanic NN: 0.9665072
```

```
cat("Precision of Titanic NN:", precision_nnwithk10, "\n")
```

```
## Precision of Titanic NN: 0.9156627
```

```
cat("Recall of Titanic NN:", reca_nnwithk10, "\n")
```

```
## Recall of Titanic NN: 1
```

```
cat("F1-Score of Titanic NN:", f1_nnwithk10, "\n")
```

```
## F1-Score of Titanic NN: 0.9559748
```